Bioaccumulation in Aquatic Organisms

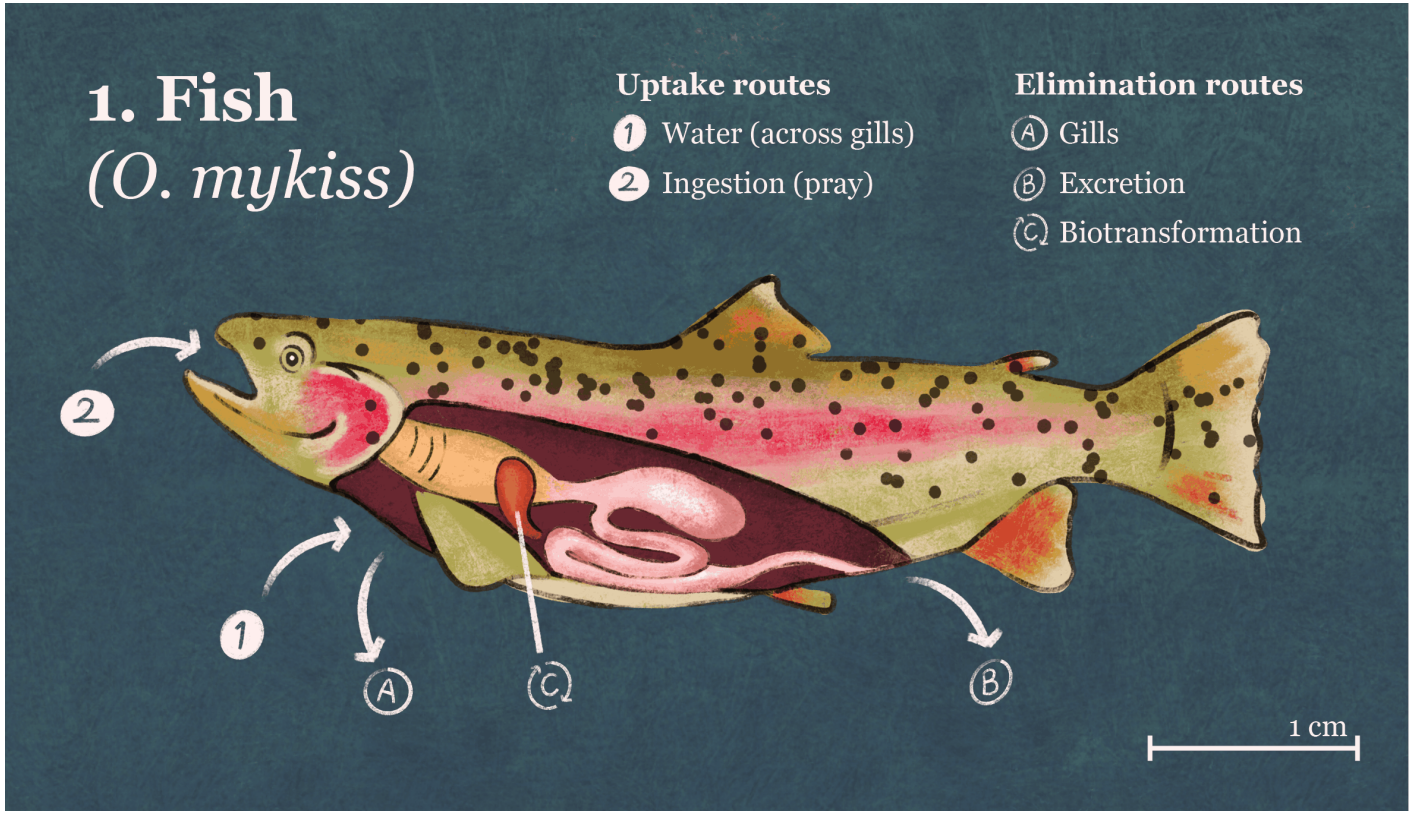
From One-Compartment Toxicokinetic Models to Real-World Complexity



Johannes Raths

Eawag: Swiss Federal Institute of Aquatic Science and Technology, Switzerland

https://orcid.org/0000-0002-3258-0893



Raths, J.; Wehrli, M.; Langer, M.; Hollender, J. Bioaccumulation in a Changing Climate: Temperature-Driven Toxicokinetics in Aquatic and Terrestrial Ectotherms. *CHIMIA* **2026**, *80* (10), 695–700. https://doi.org/10.2533/chimia.2026.695

University of Waterloo; November 12, 2026

Schedule

12th November 2026

13:00–13:15
Welcome and why bioaccumulation matters

13:15 – 13:30
Bioaccumulation – a kinetic problem

13:30–14:00
Exercise 1: The 1-compartment TK model

14:00-14:15
Plenary discussion of Exercise 1

14:15–14:45
Exercise 2: Biotransformation

14:45-15:00
Plenary discussion of Exercise 2



Course Materials [10.5281/zenodo.24121946](https://doi.org/10.5281/zenodo.24121946)

GitHub

https://github.com/HollenderGroup/Teaching_TK_Workshop_Waterloo_12.11.2026

15:00–15:15
Coffee break

15:15–15:45
Exercise 3: Multiple compartment kinetics

15:45-16:00
Plenary discussion of Exercise 3

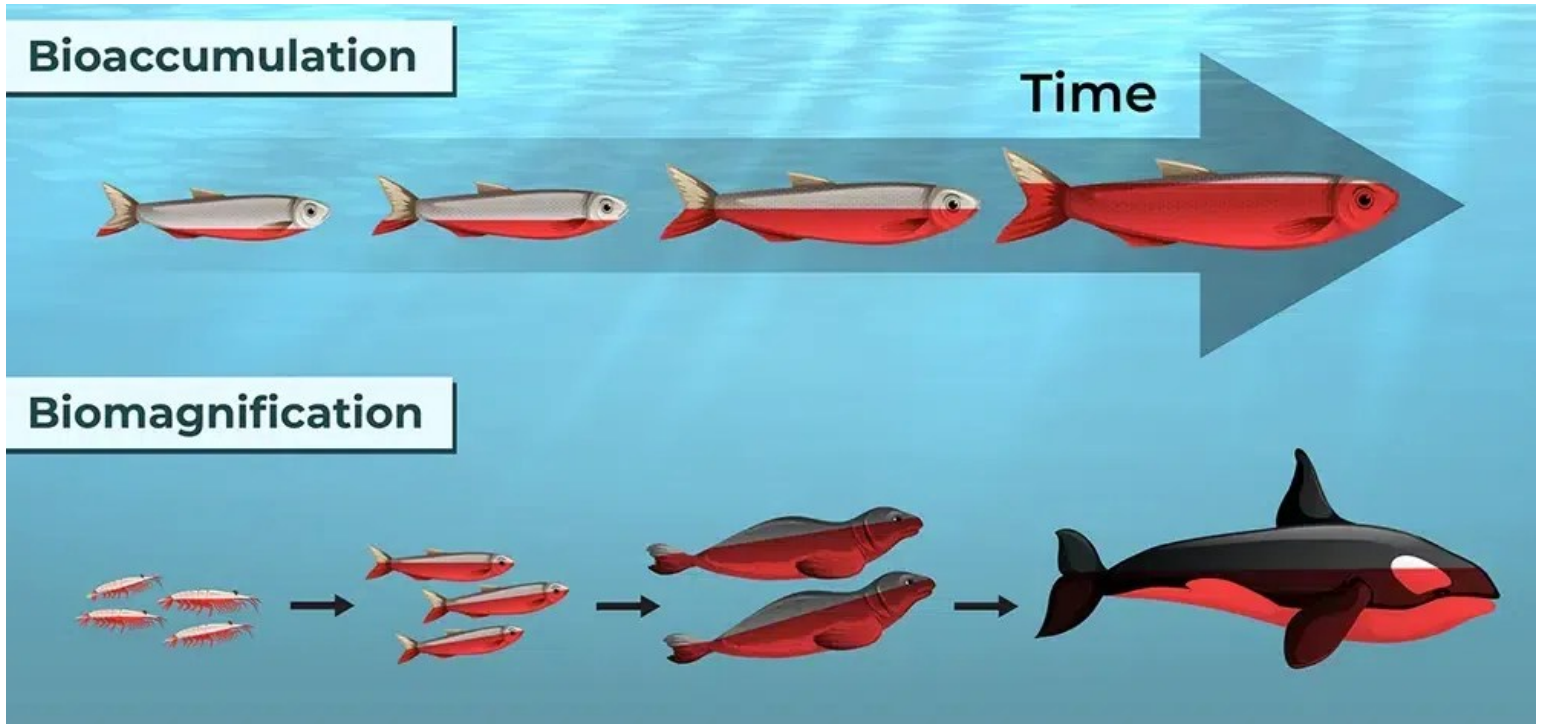
16:00–16:30
Outlook: Dietary uptake, temperature effects and others

16:30–17:00
Final discussion and Wrap-up

13:00–13:15 | Welcome and why bioaccumulation matters

What is bioaccumulation?

The definitions



https://www.geeksforgeeks.org/biology/biomagnification/

Bioaccumulation:

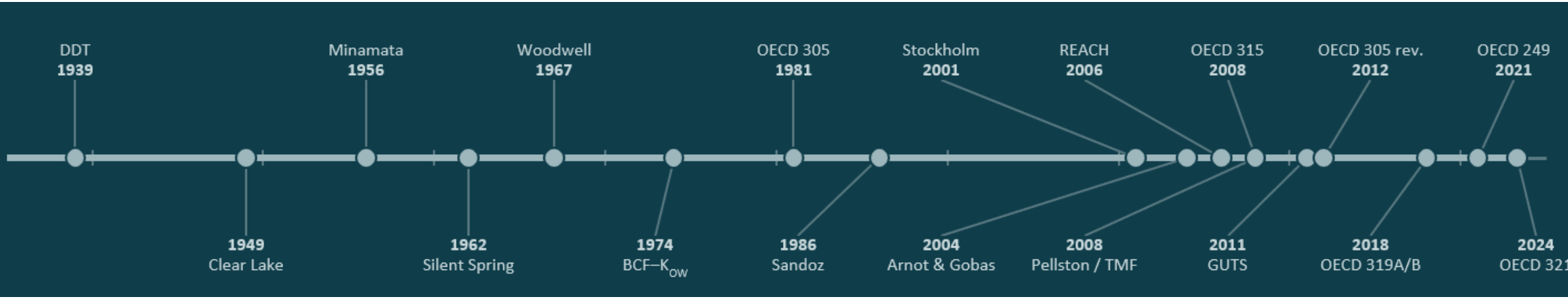
$$C_{Tissue} > C_{Medium}$$

Biomagnification:

$$C_{Tissue} > C_{Diet}$$

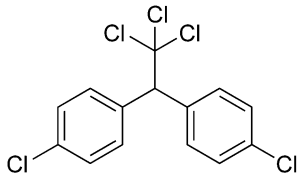
Why bioaccumulation matters

Historic and current examples of bioaccumulation



DDT

1939–1948





Paul Müller discovered DDT's insecticidal activity (Nobel Prize 1948). Its worldwide use led to the first large-scale cases of persistent, lipophilic chemicals building up in organisms.

DDT was first made in 1874 by the chemist Othmar Zeidler, but no use for it was found.



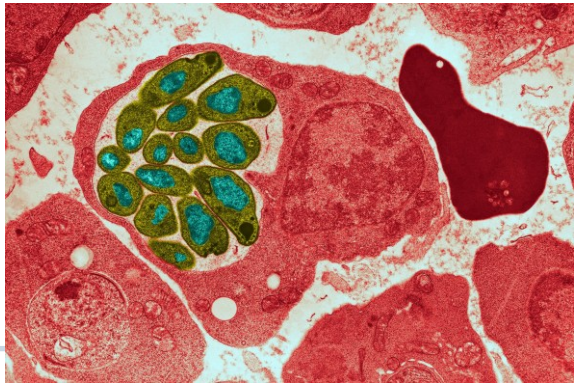
https://clickamericana.com/tag/poisons



https://culturacientifica.com/2017/12/04/historias-la-malaria-paul-muller-ddt-rachel-carson/

DDT
1939





https://www.nytimes.com/2020/10/26/health/malaria-parasites-genetics.html

Clear Lake, California

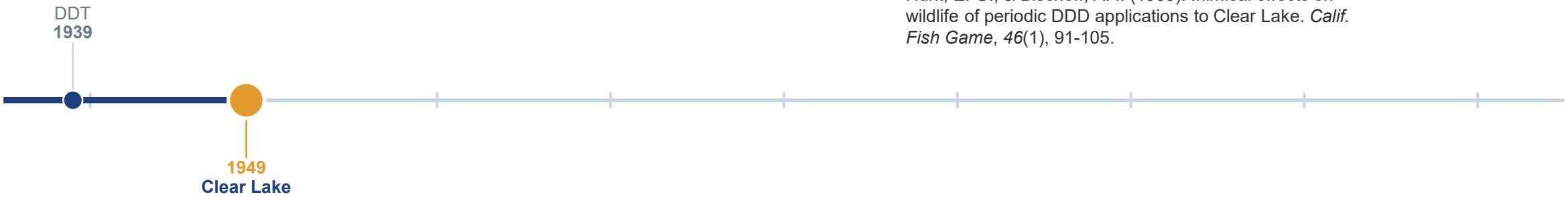
1949–1957

DDD was applied to control gnats, and western grebes later died off. Residues in grebe fat were many orders of magnitude higher than in the water (Hunt & Bischoff, 1960).
This became a textbook case of biomagnification.



https://a-z-animals.com/animals/grebe/

Hunt, E. G., & Bischoff, A. I. (1960). Inimical effects on wildlife of periodic DDD applications to Clear Lake. *Calif. Fish Game*, *46*(1), 91-105.



Minamata disease, Japan

1956

Methylmercury from Chisso's wastewater accumulated in fish and shellfish and poisoned the people who ate them.
A second outbreak followed in Niigata in 1965. It showed that a metal could biomagnify once it was transformed into an organometallic form.

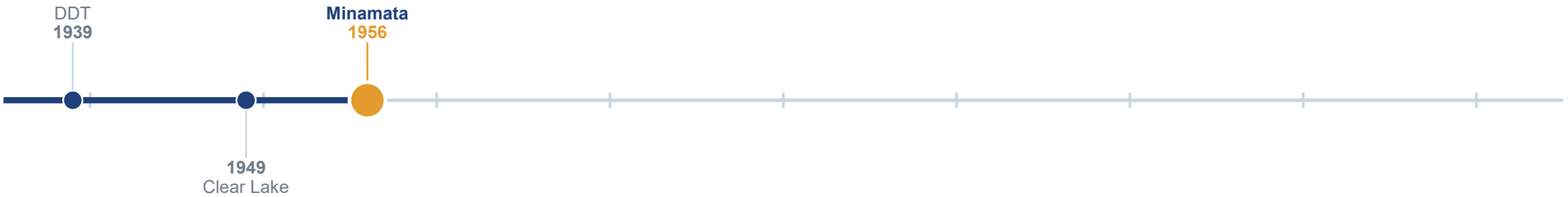


https://mainichi.jp/english/graphs/20211221/hpe/00m/0na/001000g/20211221hpe00m0na004000q

https://news.immunologic.org/p/vaccines-dont-contain-harmful-mercury

METHYLMERCURY

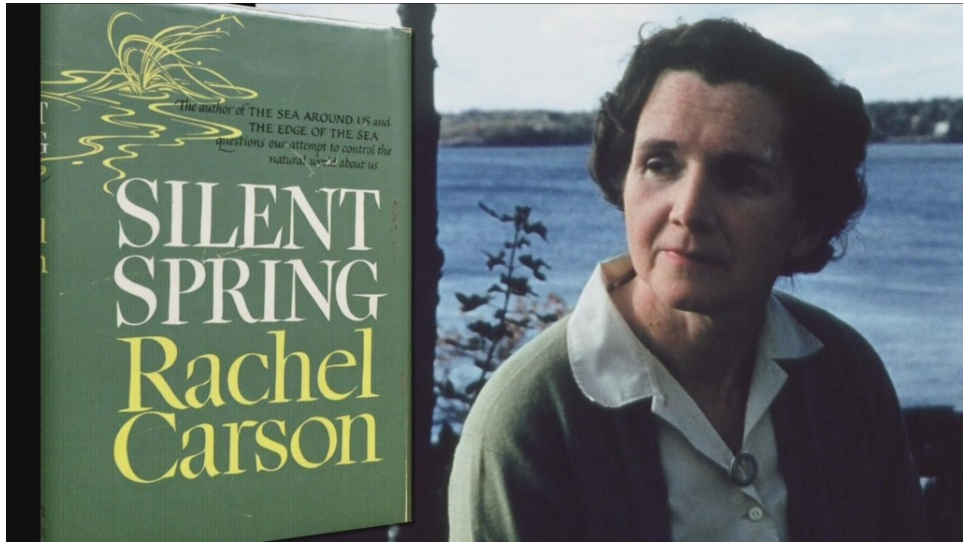




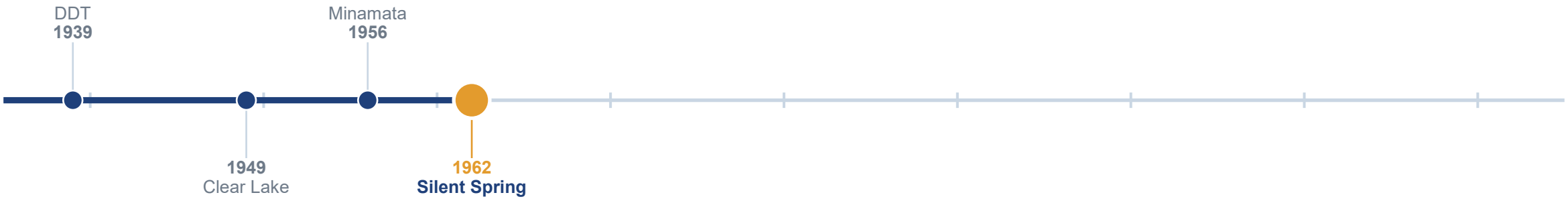
Rachel Carson's *Silent Spring*

1962

It brought food-chain accumulation of pesticides to public attention and triggered regulation.



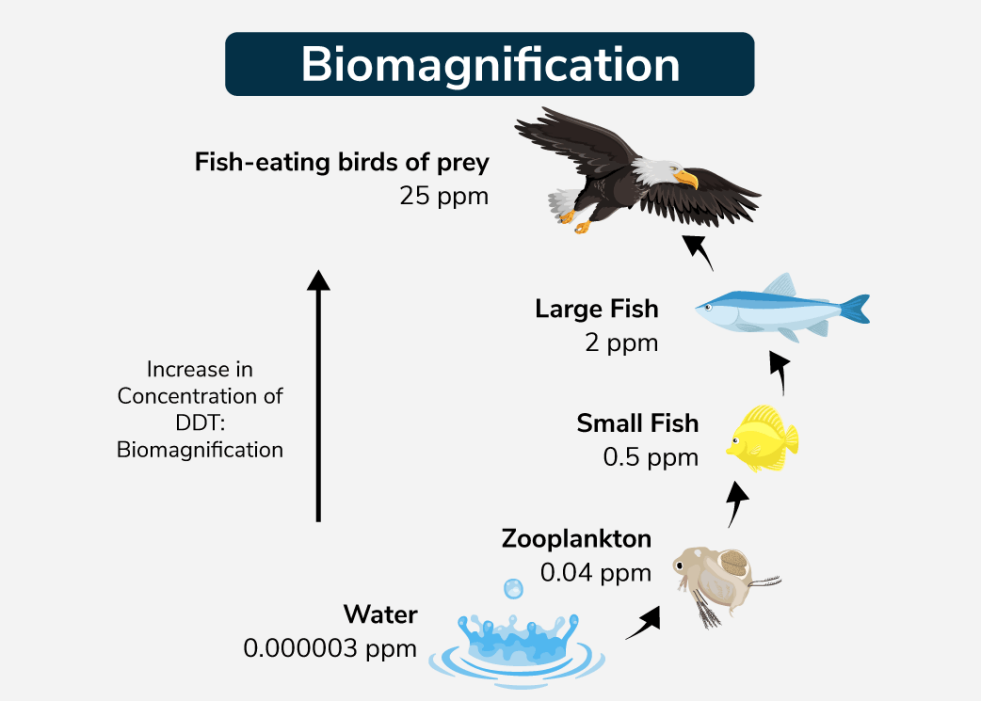
https://theenvironmentconsultant.com/2025/05/14/silent-spring-the-book-that-sparked-the-global-environmental-movement/



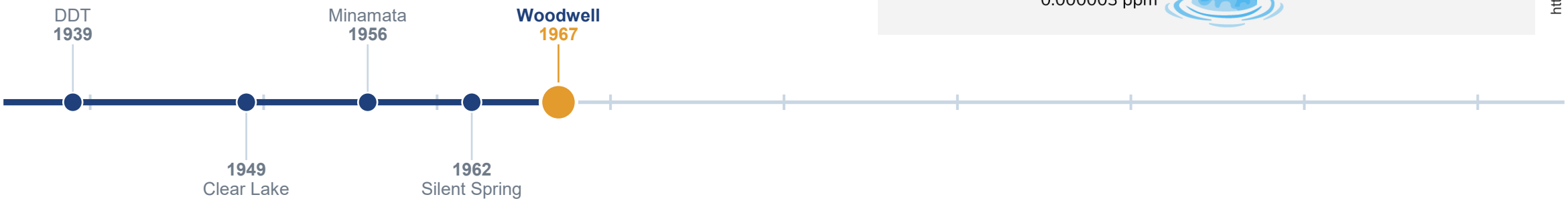
Case study: DDT Biomagnification

1967

DDT increasing up an estuarine food web.



Woodwell, G. M.; Wurster, C. F.; Isaacson, P. A. DDT Residues in an East Coast Estuary: A Case of Biological Concentration of a Persistent Insecticide. *Science* **1967**, *156* (3776), 821–824. https://doi.org/10.1126/science.156.3776.821



First QSAR approaches

1974

Correlation of the bioconcentration factor (BCF) in fish with octanol–water partition coefficient (K_{OW}).
This began quantitative structure–activity approaches to bioaccumulation.

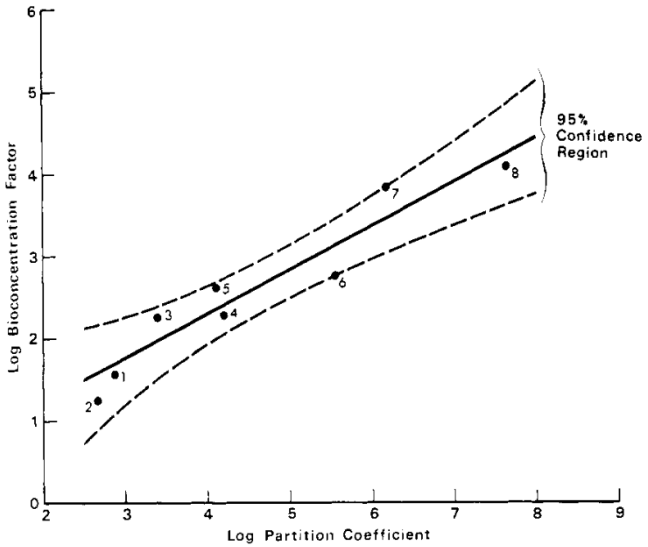
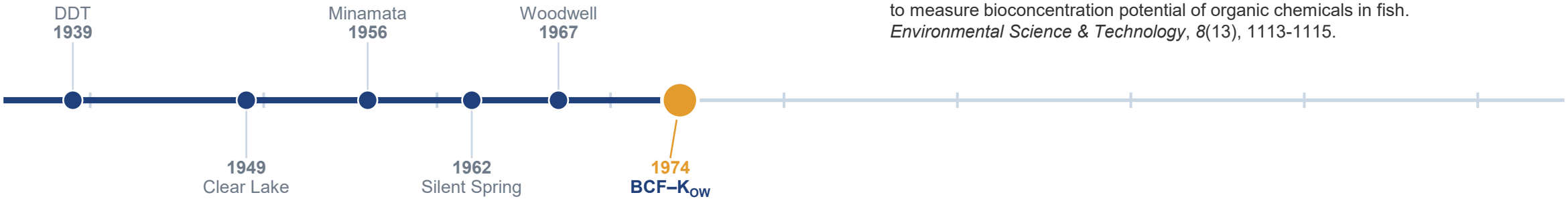


Figure 1. Linear regression between logarithms of partition coefficient and bioconcentration of various chemicals in trout muscle

Neely, W. B., Branson, D. R., & Blau, G. E. (1974). Partition coefficient to measure bioconcentration potential of organic chemicals in fish. *Environmental Science & Technology*, *8*(13), 1113-1115.



OECD Test Guideline 305

1981

First standardised fish bioconcentration test with an uptake and a depuration phase.

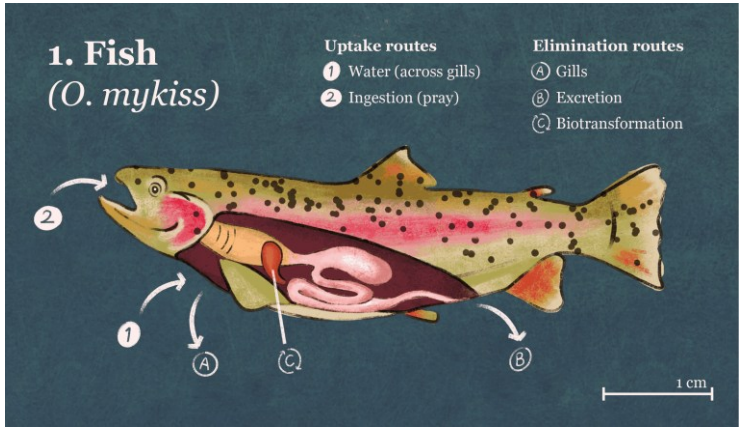
OECD. *Test No. 305: Bioaccumulation in Fish: Aqueous and Dietary Exposure*; OECD publishing, 2012.

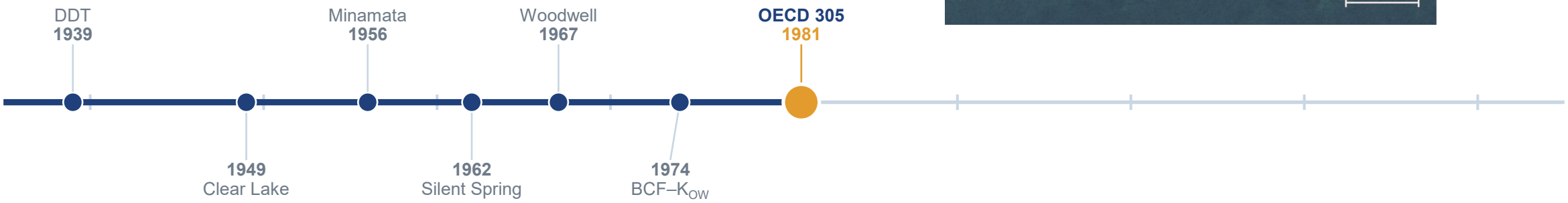
OECD/OCDE

305
Adopted:
2 October 2012

OECD GUIDELINES FOR TESTING OF CHEMICALS

Bioaccumulation in Fish: Aqueous and Dietary Exposure





Sandoz/Schweizerhalle fire, Basel

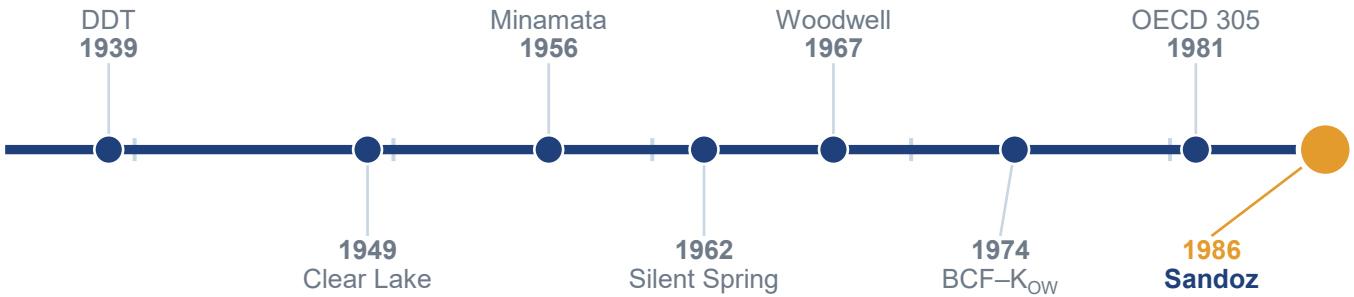
1986

Pesticides and mercury compounds were released into the Rhine.
This had a large impact on Swiss and European water research, including at Eawag.

- Organophosphates
- Metoxuron
- Organomercury fungicides
- Rhodamine B



https://www.bazonline.ch/schweizerhalle-brandort-wird-ueberwacht-871766357220



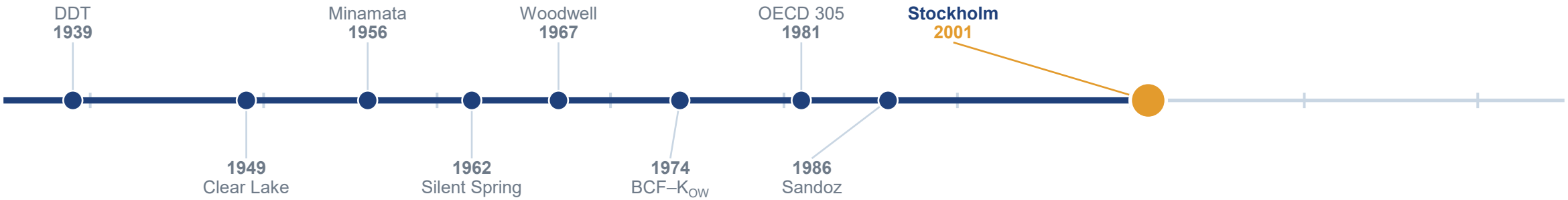


https://www.iasgyan.in/daily-current-affairs/rhine-river

Stockholm Convention (in force 2004)

2001

Definition of the bioaccumulation criterion:
BCF or bioaccumulation factor (BAF) > 5000



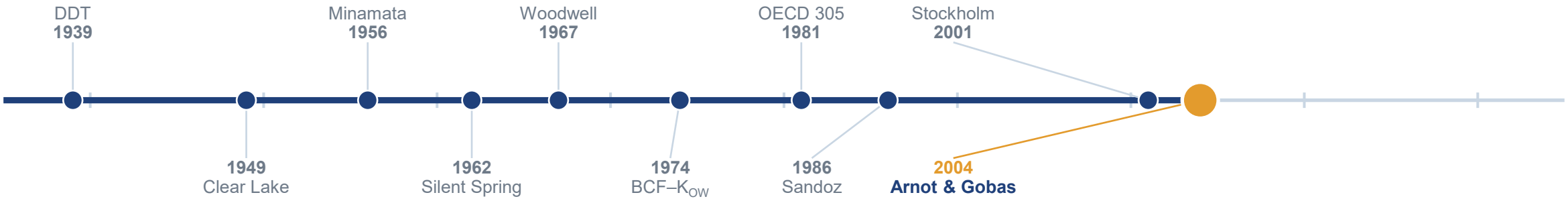
First extensive Reviews

2004–2006

Arnot & Gobas published a widely used food-web bioaccumulation model and a critical review of measured BCF/BAF data.

Arnot, J. A.; Gobas, F. A. P. C. A Food Web Bioaccumulation Model for Organic Chemicals in Aquatic Ecosystems. *Environmental Toxicology and Chemistry* **2004**, *23* (10), 2343–2355. https://doi.org/10.1897/03-438

Arnot, J. A.; Gobas, F. A. A Review of Bioconcentration Factor (BCF) and Bioaccumulation Factor (BAF) Assessments for Organic Chemicals in Aquatic Organisms. *Environ. Rev.* **2006**, *14* (4), 257–297. https://doi.org/10.1139/a06-005



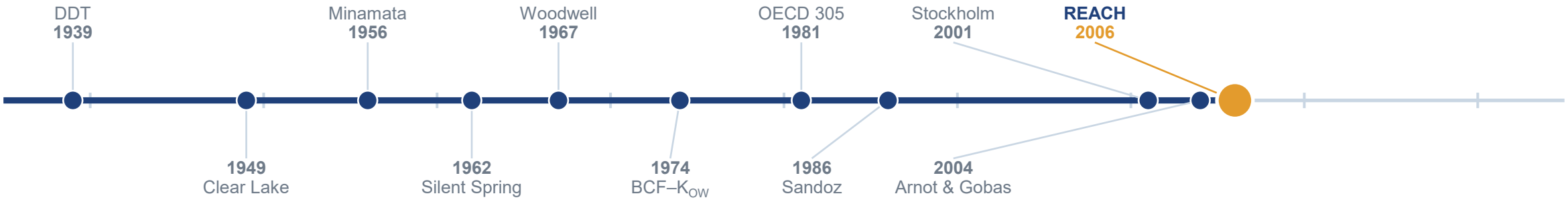
EU REACH

2006/2007

PBT/vPvB criteria (Annex XIII) defined:
Bioaccumulative (B): BCF > 2000
Very Bioaccumulative (vB): BCF > 5000.



https://www.certaintysoftware.com/eu-reach-regulation/

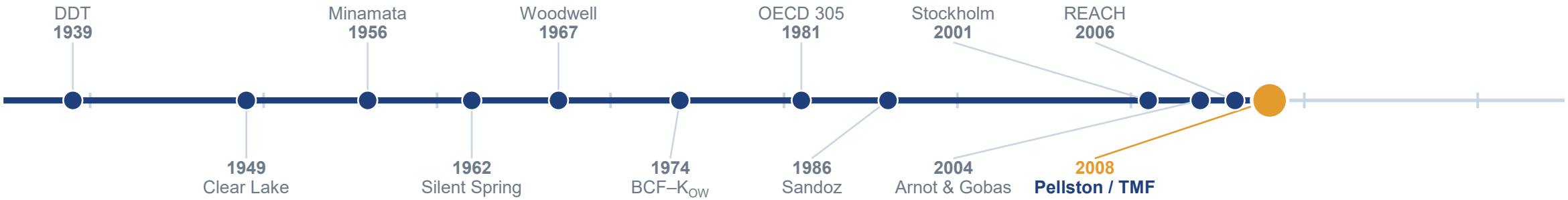


SETAC Pellston workshop

2008–2009

Bioaccumulation assessment in the field:
Trophic magnification factors (TMFs)

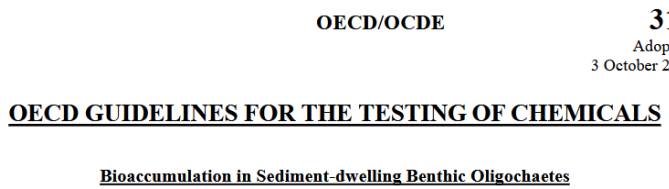
Fremlin, K. M.; Elliott, J. E.; Gobas, F. A. P. C. Guidance for Measuring and Evaluating Biomagnification Factors and Trophic Magnification Factors of Difficult Substances: Application to Decabromodiphenylethane. *Integr Environ Assess Manag* **2025**, *21* (2), 263–278. https://doi.org/10.1093/inteam/vjae025



OECD TG 315 / 317

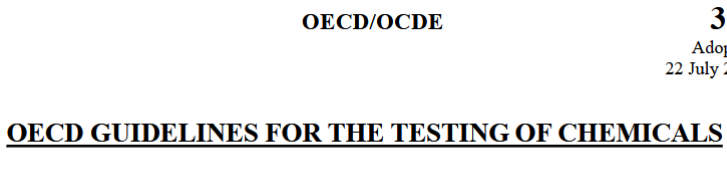
2008-2010

Bioaccumulation test with sediment-dwelling oligochaete worms and earthworms in soil.


OECD/OCDE 315 Adopted: 3 October 2008

OECD GUIDELINES FOR THE TESTING OF CHEMICALS

Bioaccumulation in Sediment-dwelling Benthic Oligochaetes

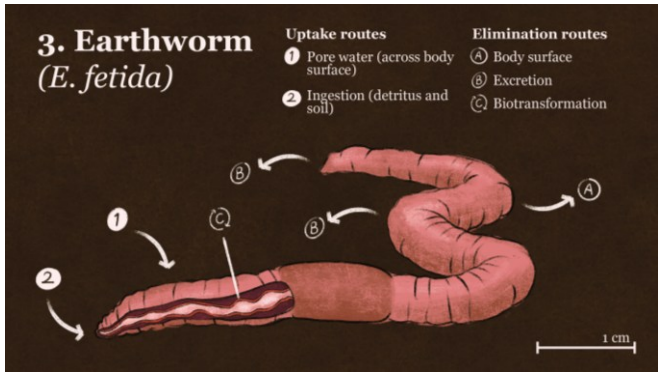

OECD/OCDE 317 Adopted: 22 July 2010

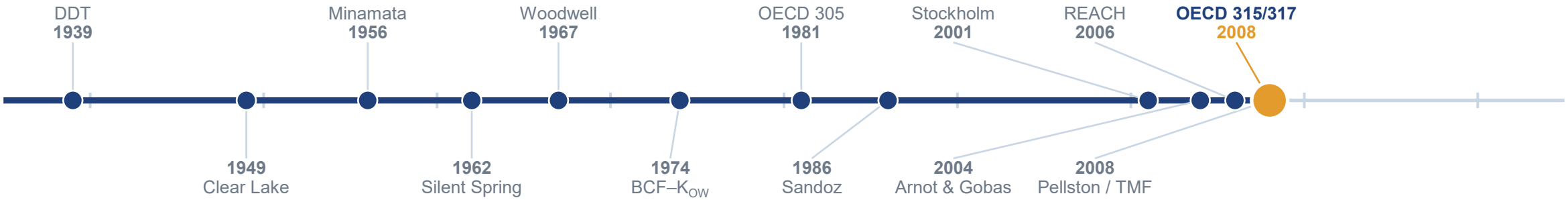
OECD GUIDELINES FOR THE TESTING OF CHEMICALS

Bioaccumulation in Terrestrial Oligochaetes

OECD. *Test No. 315: Bioaccumulation in Sediment-Dwelling Benthic Oligochaetes*; OECD Guidelines for the Testing of Chemicals, Section 3; OECD, 2008. https://doi.org/10.1787/9789264067516-en

OECD. *Test No. 317: Bioaccumulation in Terrestrial Oligochaetes*; OECD Guidelines for the Testing of Chemicals, Section 3; OECD, 2010. https://doi.org/10.1787/9789264090934-en

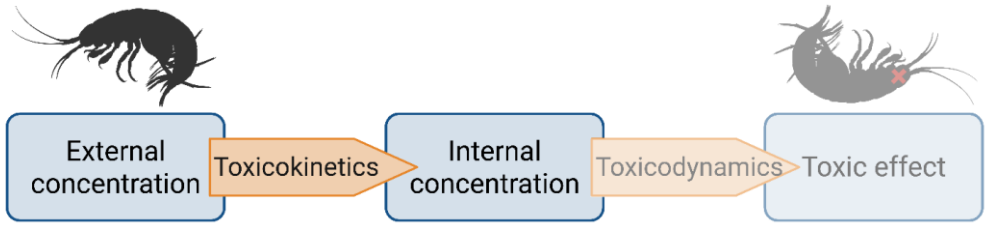




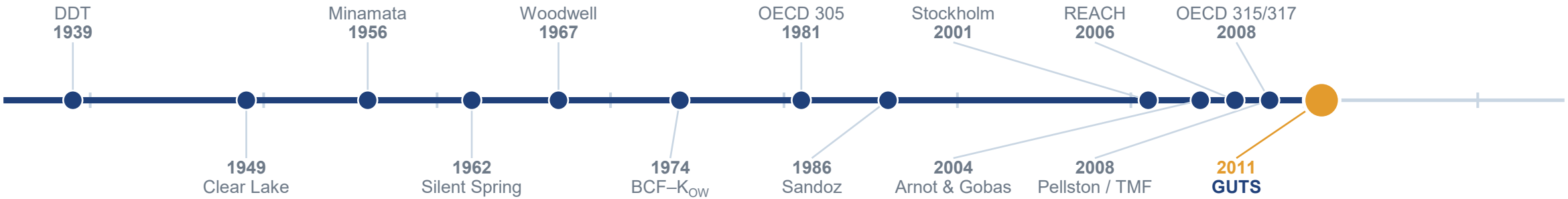
GUTS

2011

The General Unified Threshold model of Survival unified TK-TD survival models.
It is now part of EFSA's guidance for pesticide risk assessment.



Jager, T.; Albert, C.; Preuss, T. G.; Ashauer, R. General Unified Threshold Model of Survival - a Toxicokinetic-Toxicodynamic Framework for Ecotoxicology. *Environ. Sci. Technol.* **2011**, *45* (7), 2529–2540. https://doi.org/10.1021/es103092a



OECD 305 revision

2012

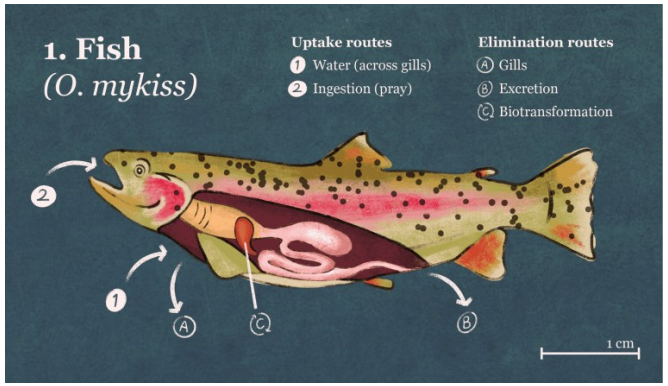
It added a dietary exposure test for very hydrophobic chemicals and a reduced-fish (“minimized”) design.

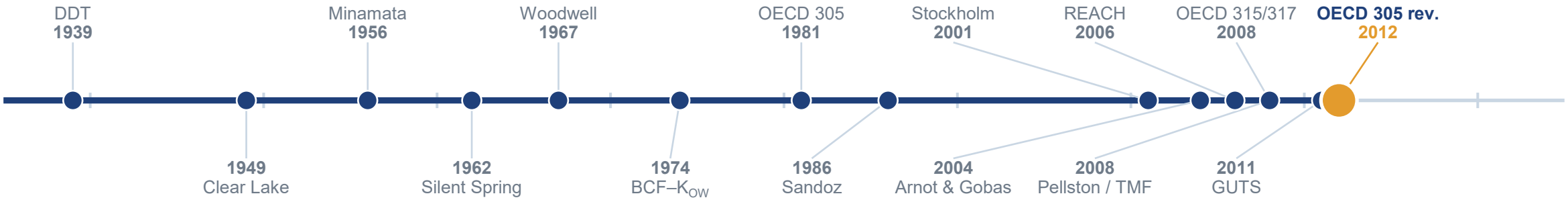
OECD/OCDE

305
Adopted:
2 October 2012

OECD GUIDELINES FOR TESTING OF CHEMICALS

Bioaccumulation in Fish: Aqueous and Dietary Exposure

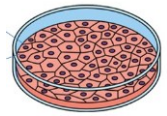




OECD TG 319A/B

2018

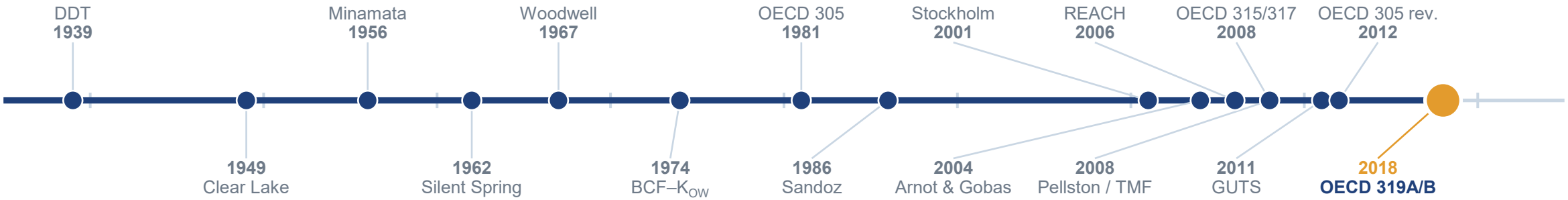
In vitro biotransformation tests using rainbow trout liver cells and liver S9 fractions.
Combined with in vitro–in vivo extrapolation (IVIVE), they help predict BCFs with fewer fish.



Test No. 319A: Determination of in vitro intrinsic clearance using cryopreserved rainbow trout hepatocytes (RT-HEP)



Test No. 319B: Determination of in vitro intrinsic clearance using rainbow trout liver S9 sub-cellular fraction (RT-S9)



OECD TG 249

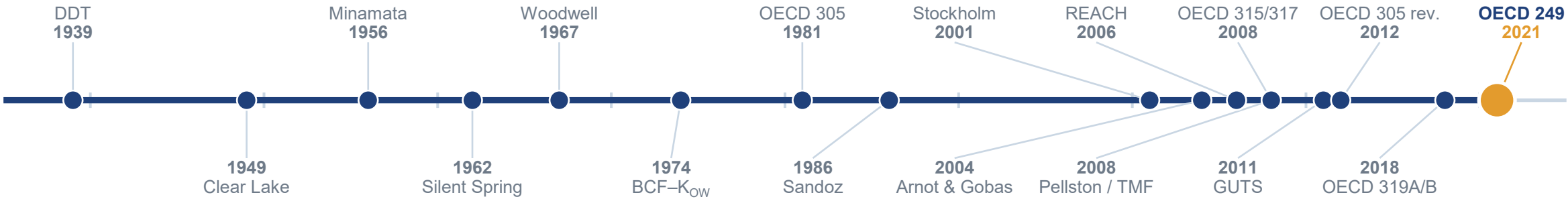
2021

The RTgill-W1 fish gill cell line assay (developed at Eawa). It is a 3R alternative for fish acute toxicity (not bioaccumulation)

But:
Such assays can also be used for bioaccumulation estimation (i.e. Stadnicka-Michalak et al., 2018)

Test No. 249: Fish Cell Line Acute Toxicity - The RTgill-W1 cell line assay

Stadnicka-Michalak, J.; Weiss, F. T.; Fischer, M.; Tanneberger, K.; Schirmer, K. Biotransformation of Benzo[a]Pyrene by Three Rainbow Trout (*Onchorhynchus Mykiss*) Cell Lines and Extrapolation To Derive a Fish Bioconcentration Factor. *Environ. Sci. Technol.* **2018**, *52* (5), 3091–3100. https://doi.org/10.1021/acs.est.7b04548



OECD TG 321

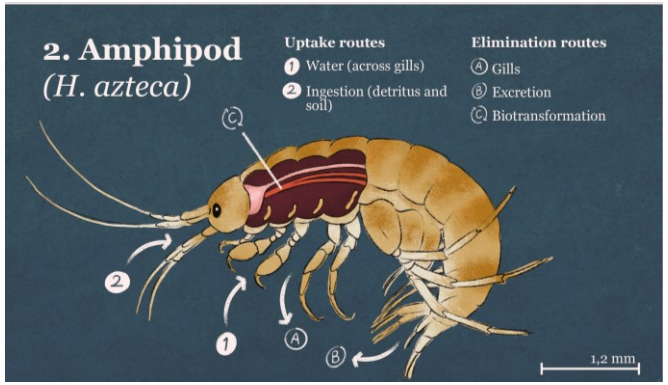
2024

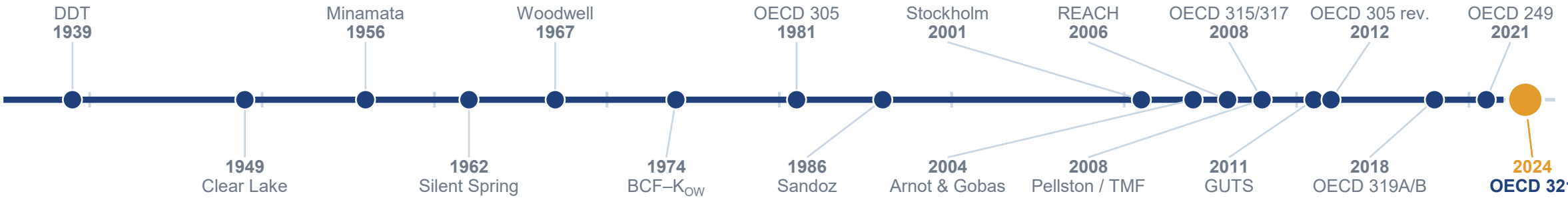
A bioconcentration test with the freshwater amphipod *Hyalella azteca*, an invertebrate alternative to fish tests.

OECD. *Test No. 321: Hyallela Azteca Bioconcentration Test (HYBIT)*; OECD Guidelines for the Testing of Chemicals, Section 3; OECD, 2024. https://doi.org/10.1787/8ac30c4e-en

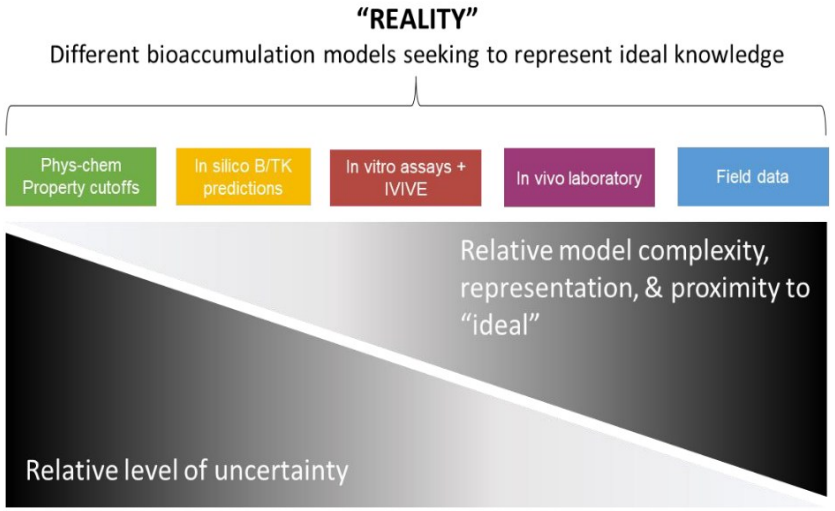
Test Guideline No. 321

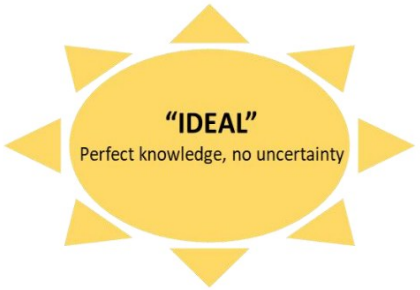
Hyalella Azteca Bioconcentration Test (HYBIT)





Bioaccumulation assessment framework





OECD. Case Study on the Use of Integrated Approaches for Testing and Assessment (IATA) for Bioaccumulation - Ninth Review Cycle (2023). *OECD Series on Testing and Assessment* **2024**. https://doi.org/10.1787/8bc8ad6f-en

Taxa	Data type(s)	Parameter / metric
Aquatic & terrestrial	Phys-Chem (measured & *in silico*)	Molecular weight
		K_{OW}
		K_{OA} (terrestrial homeotherms only)
		Water solubility
		Vapour pressure
		Henry's Law Constant
		Air-water partition ratio (K_{AW})
		Solubility in octanol (S_O)
		Biopartitioning parameters
		pKa (IOC* only)
	Biotransformation	*In vivo* biotransformation and elimination rate constants and half-life data
		In silico biotransformation and elimination rate constants and half-life data
		In vitro biotransformation and elimination rate constants and half-life data
		In vitro – in vivo extrapolation
		Biotransformation in additional species / taxa
	Bioaccumulation	*In silico* bioaccumulation models
		Laboratory BCF data
		Laboratory BMF data (Aquatic)
		Laboratory TK and BMF data (Mammals)
		Field BAF
		Field BMF
		Field TMF

Regulatory requirement:
OECD TG 305 (default)
OECD TG 321 (alternative, EU)

Supporting evidence only:
OECD TG 315/317
OECD TG 319A/B

Regulatory cut-off criteria

Consideration of cut-off criteria for diflufenican according to Annex II of Regulation (EC) No 1107/2009 of the European Parliament and of the Council

Properties		Conclusion[a]
CMR	Carcinogenicity (C)	Diflufenican is not considered to be a carcinogen (category 1A or 1B) according to point 3.6.3 of Annex II of Regulation (EC) No 1107/2009, based on the harmonised classification (Regulation (EC) No 1272/2008 – ATP17)
	Mutagenicity (M)	Diflufenican is not considered to be a mutagen according to point 3.6.2 of Annex II of Regulation (EC) No 1107/2009, based on the harmonised classification (Regulation (EC) No 1272/2008 – ATP17)
	Toxic for Reproduction (R)	Diflufenican is not considered to be toxic for reproduction (Category 1A or 1B) according to point 3.6.4 of Annex II of Regulation (EC) No 1107/2009, based on the harmonised classification (Regulation (EC) No 1272/2008 – ATP17)
Endocrine disrupting properties		The endocrine disruption properties of diflufenican according to points 3.6.5 and 3.8.2 of Annex II of Regulation No 1107/2009, as amended by Commission Regulation (EU) 2018/605 are not met.
POP	**Persistence** **Bioaccumulation** **Long-range transport**	Diflufenican is not considered to be a persistent organic pollutant (POP) according to point 3.7.1 of Annex II of Regulation (EC) 1107/2009.
PBT	**Persistence** **Bioaccumulation** **Toxicity**	Diflufenican is not considered to be a persistent, bioaccumulative and toxic (PBT) substance according to point 3.7.2 of Annex II of Regulation (EC) 1107/2009.
vPvB	**Persistence** **Bioaccumulation**	Diflufenican not considered to be a very persistent, very bioaccumulative substance according to point 3.7.3 of Annex II of Regulation (EC) 1107/2009.

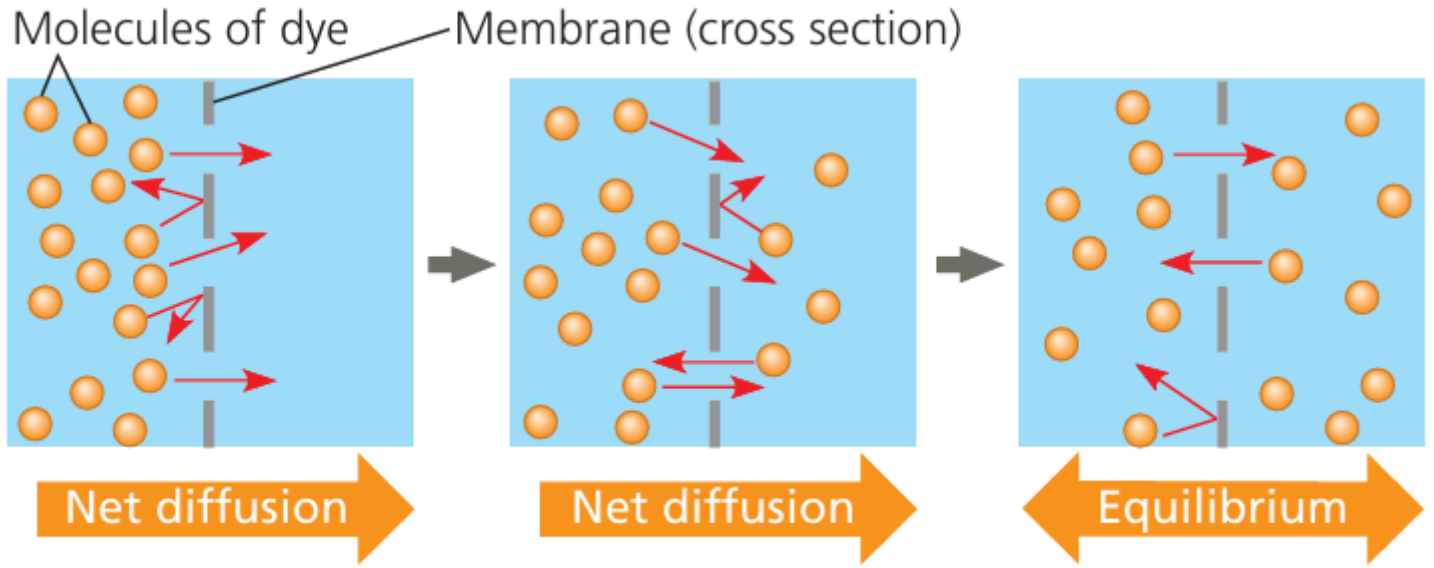
[a]Origin of data to be included where applicable (e.g. EFSA, ECHA RAC, Regulation).

Europe:
B = BCF >2000
vB = BCF >5000

13:15 – 13:30 | Bioaccumulation – a kinetic problem

Bioaccumulation – a kinetic problem

Diffusion



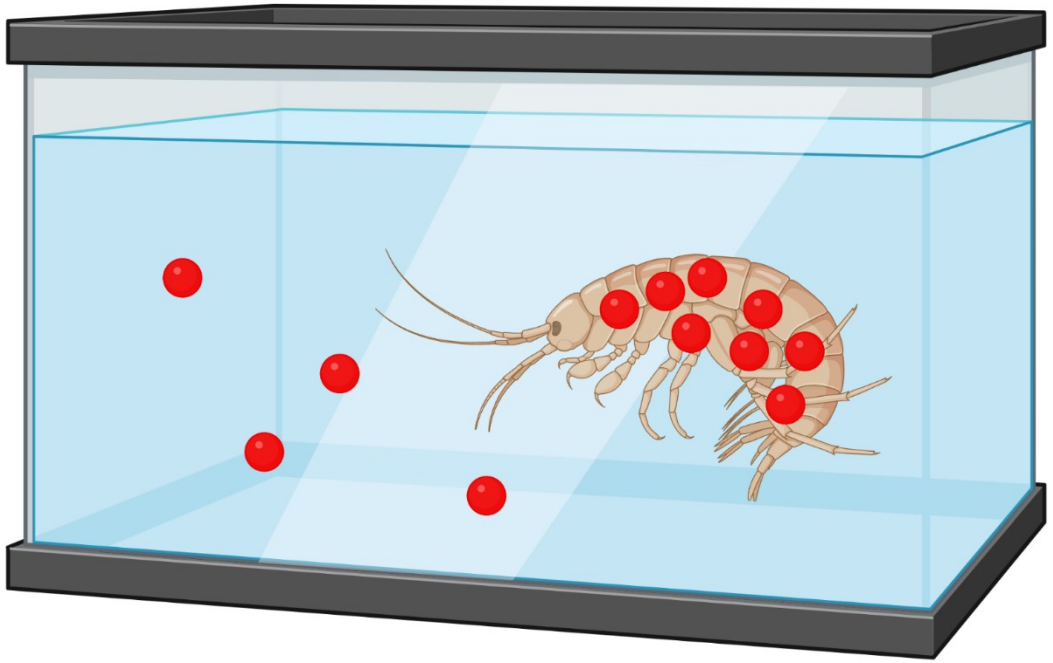
Diffusion of one solute.

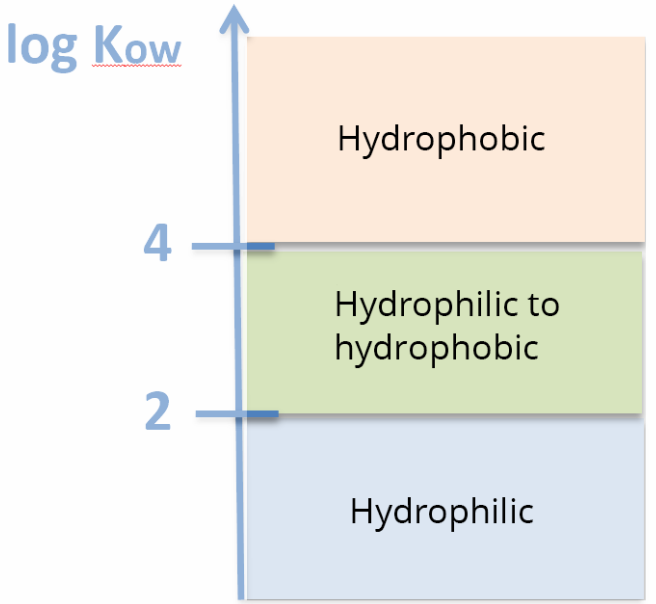
(or Steady State for living organisms)

https://www.iitianacademy.com/ib-myp-programme/ib-myp-4-5-biology-revision-resources/ib-myp-4-5-biology-diffusion-study-notes/

Bioaccumulation – a kinetic problem

Partitioning

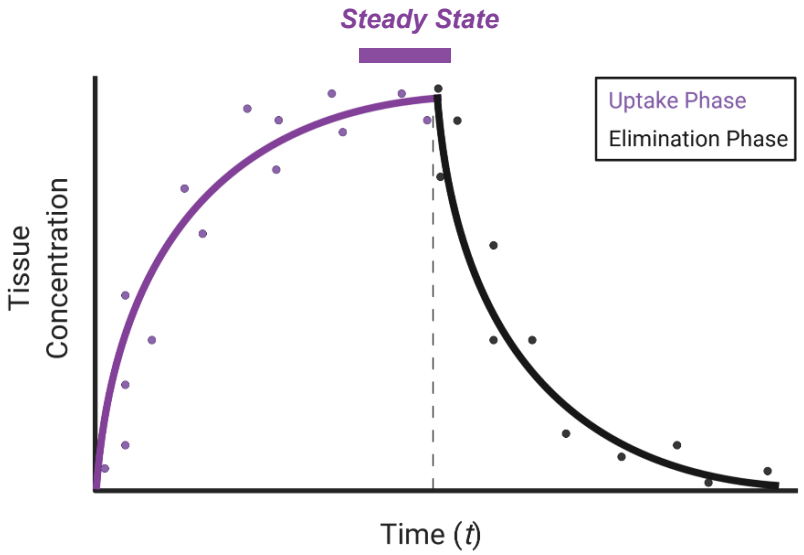


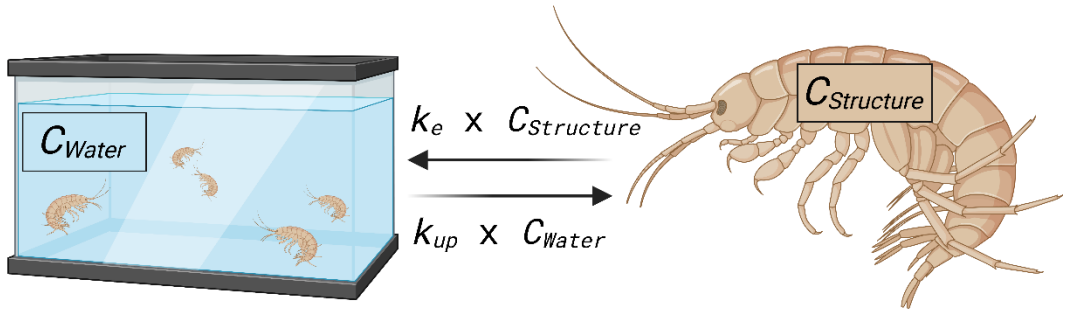


Bioaccumulation – a kinetic problem

One-compartment model

$$BCF = \frac{C_{tissue,ss}}{C_{water,ss}} = \frac{k_u}{k_e}$$





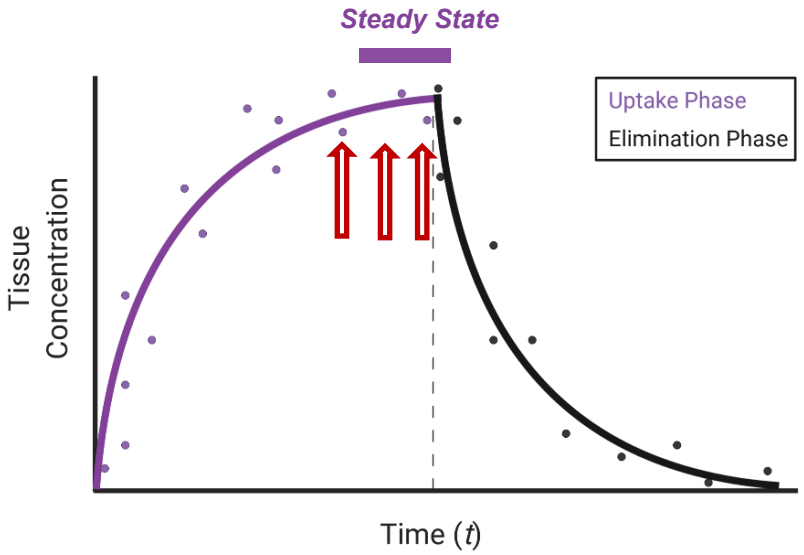
$$\frac{dC_{Structure}(t)}{dt} = k_u \cdot C_{water}(t) - k_e \cdot C_{Structure}(t)$$

Bioaccumulation – a kinetic problem

Sampling procedure

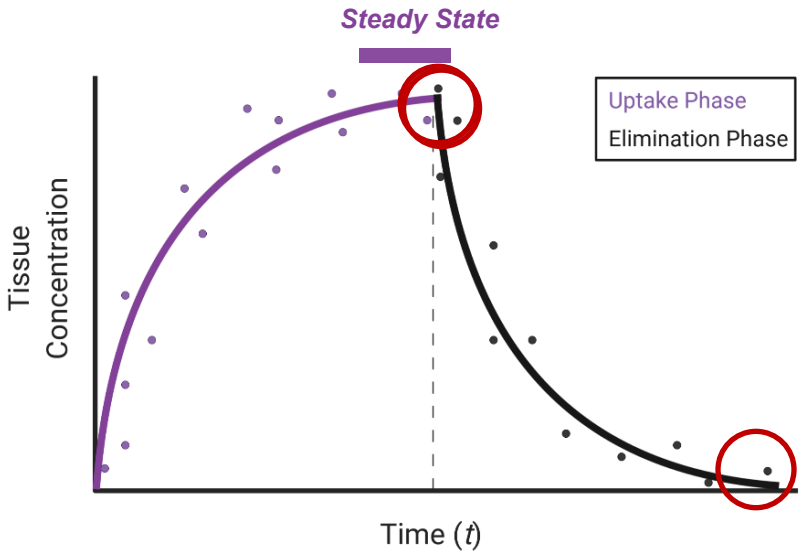
Carter, L. J.; Ashauer, R.; Ryan, J. J.; Boxall, A. B. A. Minimised Bioconcentration Tests: A Useful Tool for Assessing Chemical Uptake into Terrestrial and Aquatic Invertebrates? *Environ. Sci. Technol.* **2014**, *48* (22), 13497–13503. https://doi.org/10.1021/es5031992

OECD 305 Standard



Limitations:
- Uses a lot of animals

OECD 305 Minimized



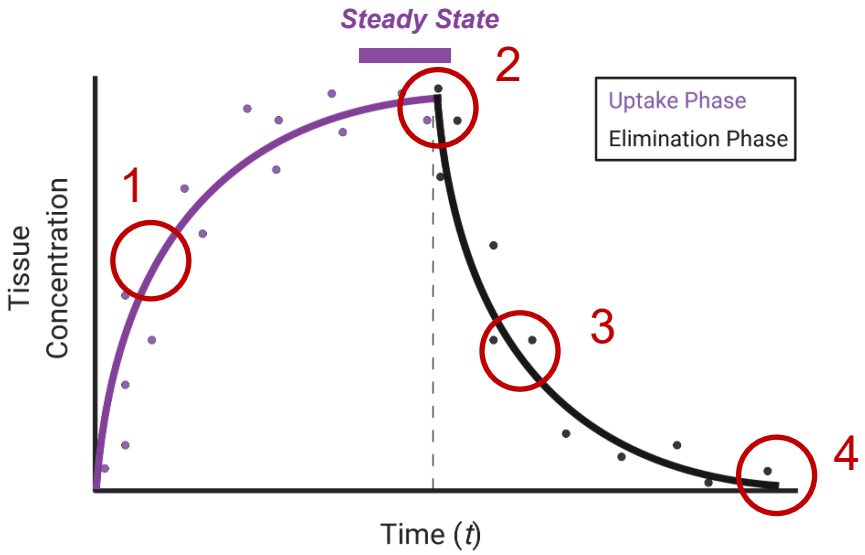
Limitations:
- No steady state confirmation
- How to interpret elimination??

Bioaccumulation – a kinetic problem

Sampling procedure

From laboratory to mesocosm: Temperature and species dependent toxicokinetics of the fluorinated herbicide Diflufenican (Sumbmitted, ES&T Toxicology)

Johannes Raths[1,2]*, Clemens Stäuble[1,2], Markus Hermann[1], Juliane Hollender[1,2]

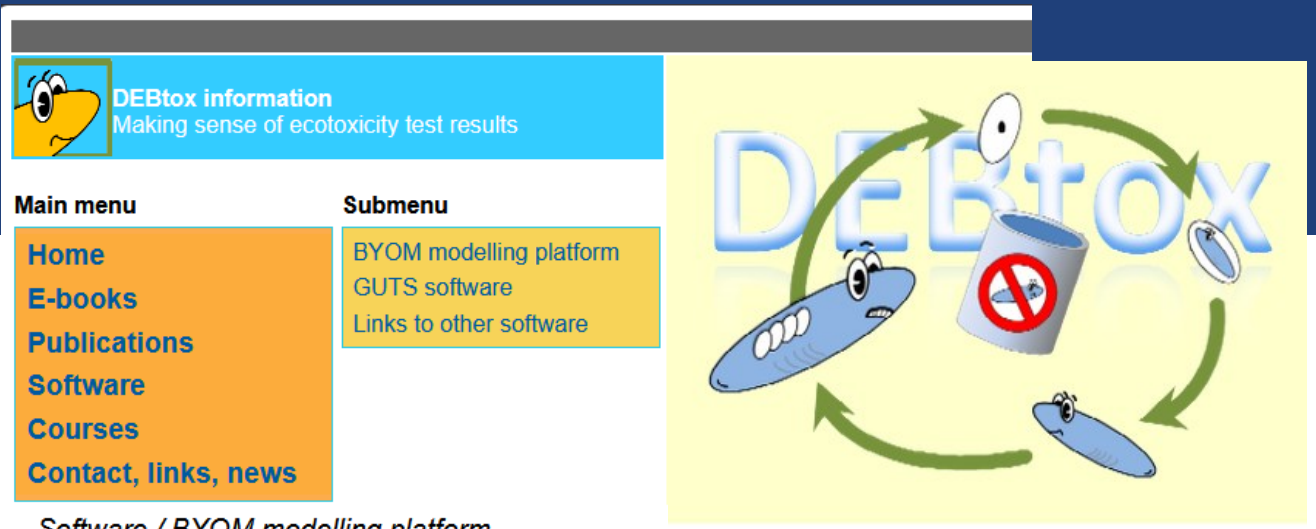


1. Confirms Steady State & provides k_e
2. Kind of necessary

3. Provides k_e & check for multi-compartment
4. Check for multi-compartment

Limitations:
- Finding good estimates ~dt50

13:30–14:00 | Exercise 1:
The standard 1-compartment TK model



https://www.debtox.info/byom.html

Wehrli, M.; Raths, J. O.; de Courten, A.; Maccagnan, A.; Fenner, K.; Junghans, M.; Renaud, M.; Langer, M.
Pesticides Never Walk Alone: Mixture Toxicity and Toxicokinetics of Five Formulated Pesticide Products in the Soil Arthropod *Folsomia Candida*. Social Science Research Network: Rochester, NY August 27, 2026.
https://doi.org/10.2139/ssrn.6832346

BYOM flexible model platform (2012-2025)

MATLAB

https://github.com/HollenderGroup/Bioaccumulation_TK-metabolism

R Studio®

https://github.com/HollenderGroup/Bioaccumulation_TK-metabolism_arrhenius

Dataset

12 compounds x 2 species x 4 temperatures

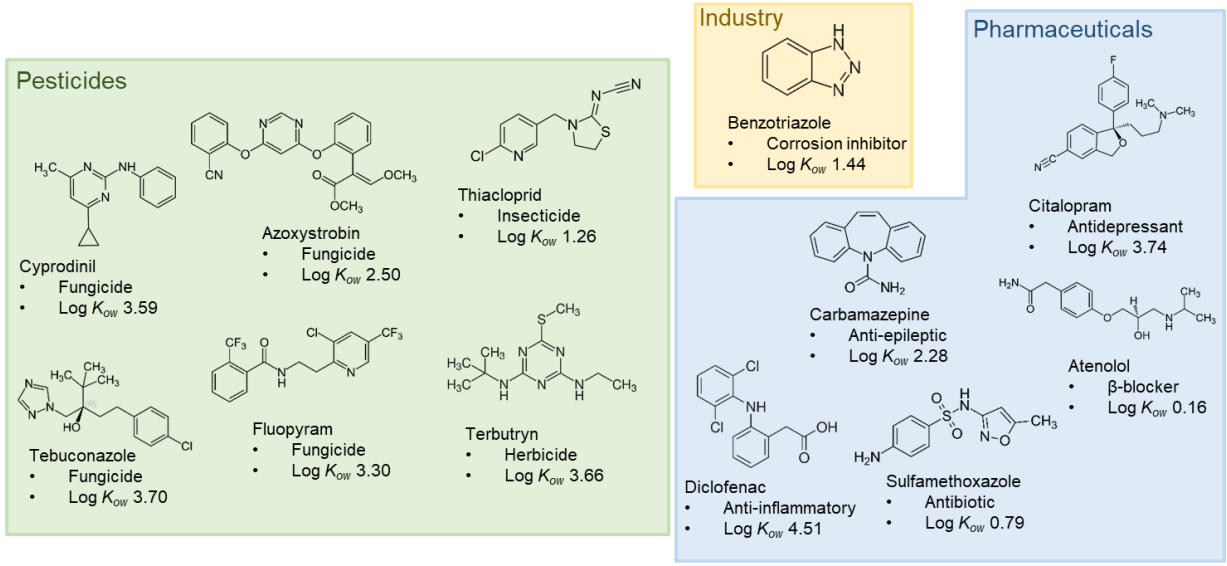
Received: 25 August 2022 | Revised: 7 November 2022 | Accepted: 18 November 2022
DOI: 10.1111/gcb.16542

Global Change Biology WILEY

RESEARCH ARTICLE

Speed it up: How temperature drives toxicokinetics of organic contaminants in freshwater amphipods

Johannes Raths[1,2] | Vid Švara[3,4] | Benedikt Lauper[1,2] | Qiuguo Fu[1] | Juliane Hollender[1,2]



Task:
Choose your favorite compound and species
→ Run the model!



G. pulex



H. azteca

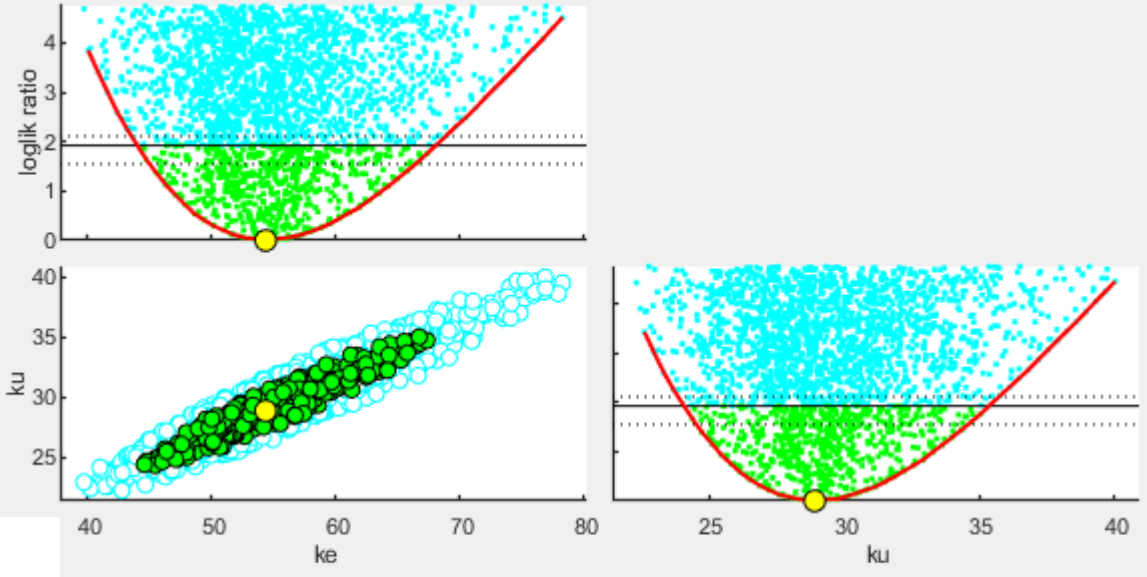
Individual TK experiments at four different temperatures



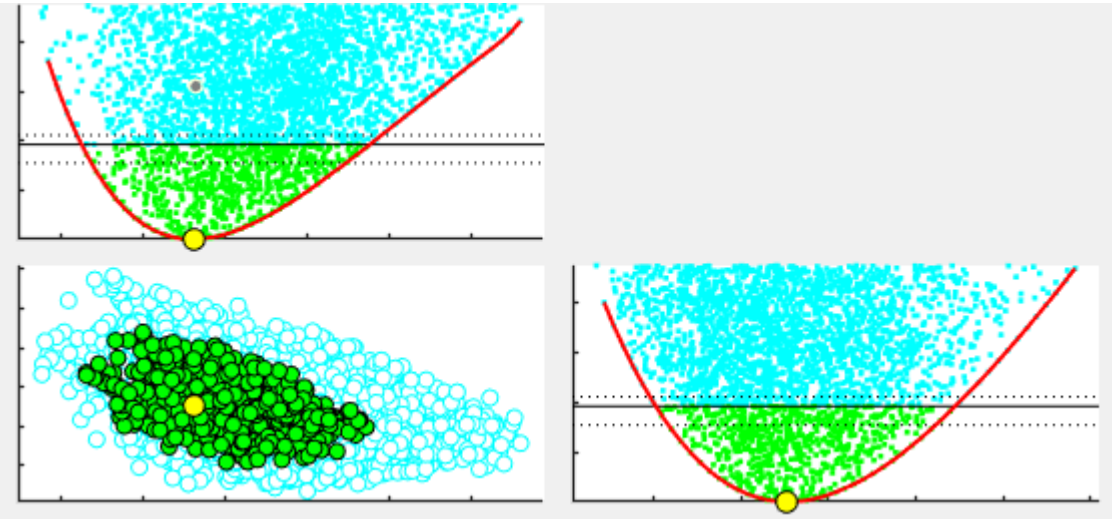
14:00-14:15 | Plenary discussion of Exercise 1

Parameter Space Explorer

How to read and interpret



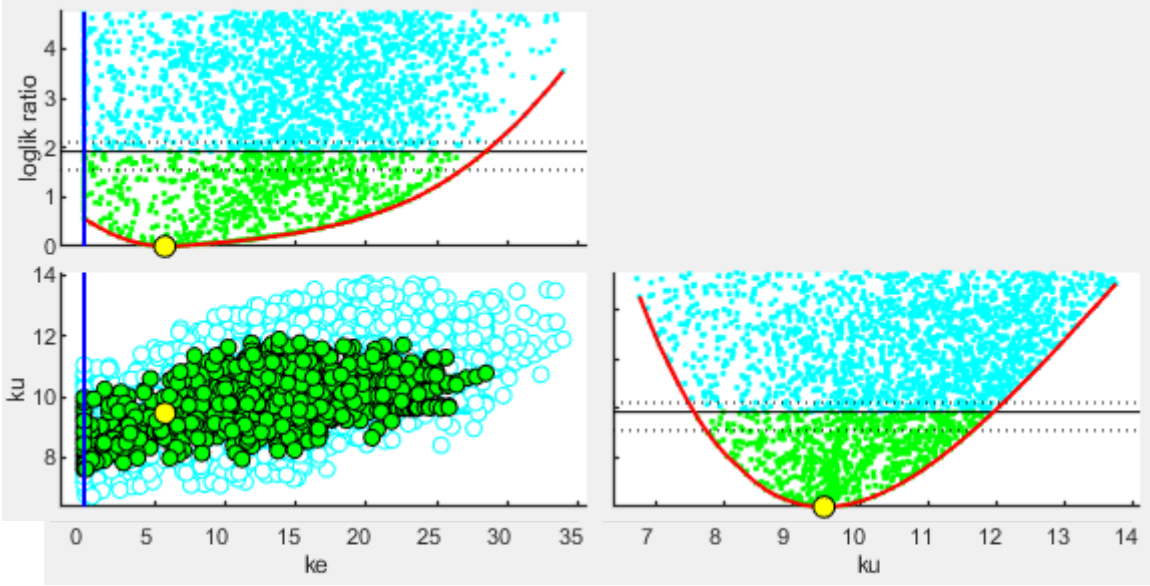
Parameters (k_e, k_u) well defined
Slight dependency



Less dependency

Parameter Space Explorer

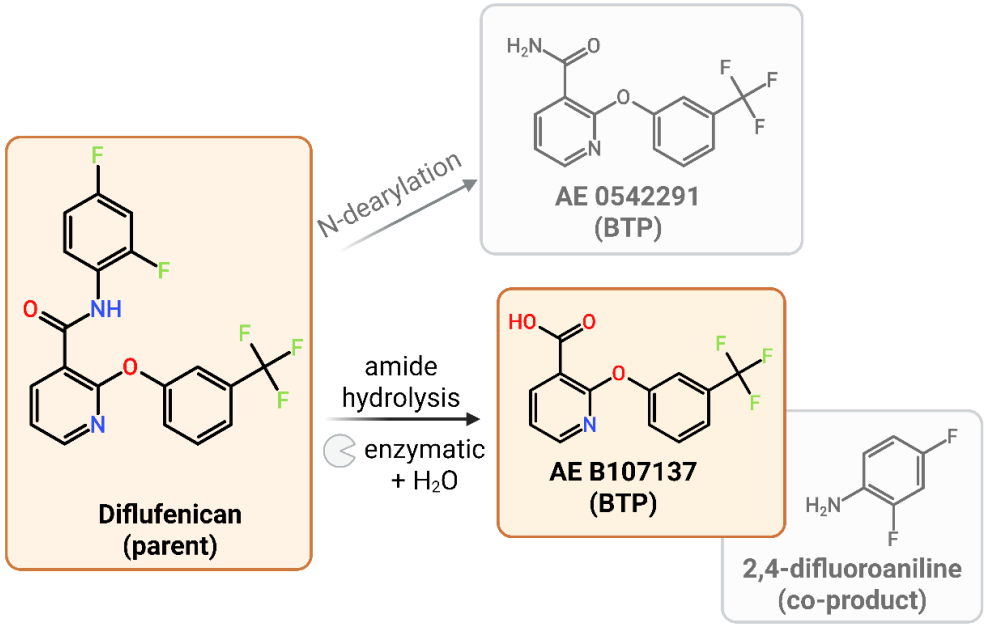
How to read and interpret



k_e meets lower boundary

14:15–14:45 | Exercise 2: Biotransformation

Biotransformation
Diflufenican



Species specific transformation

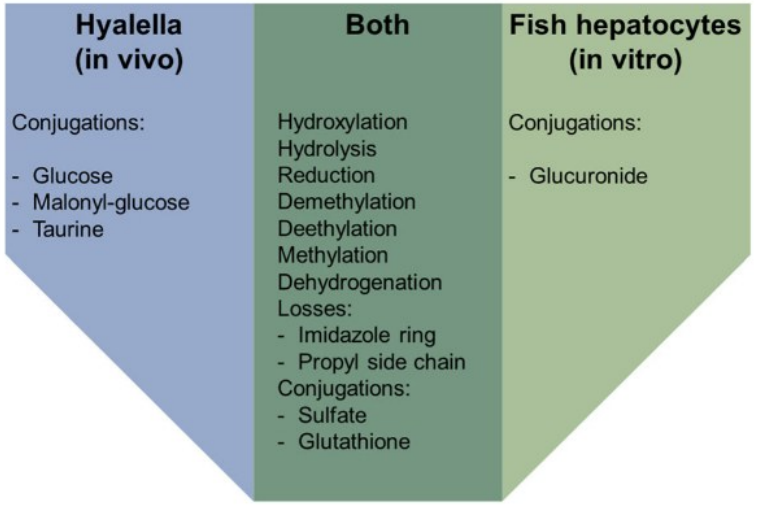


FIGURE 6: Metabolite classes of the 5 test substances detected in either *Hyalella azteca* only, rainbow trout hepatocytes only, or both sample types.

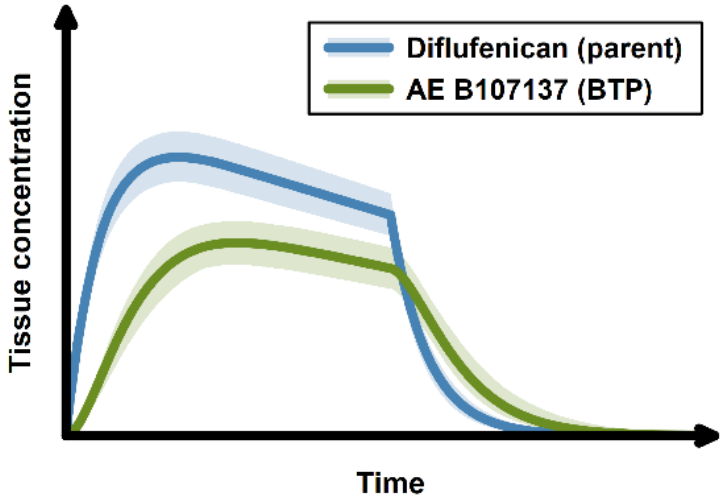
Kosfeld, V.; Fu, Q.; Ebersbach, I.; Esser, D.; Schauerte, A.; Bischof, I.; Hollender, J.; Schlechtriem, C.
Comparison of Alternative Methods for Bioaccumulation Assessment: Scope and Limitations of In Vitro Depletion Assays with Rainbow Trout and Bioconcentration Tests in the Freshwater Amphipod Hyalella Azteca. *Environmental Toxicology and Chemistry* **2020**, *39* (9), 1813–1825. https://doi.org/10.1002/etc.4791

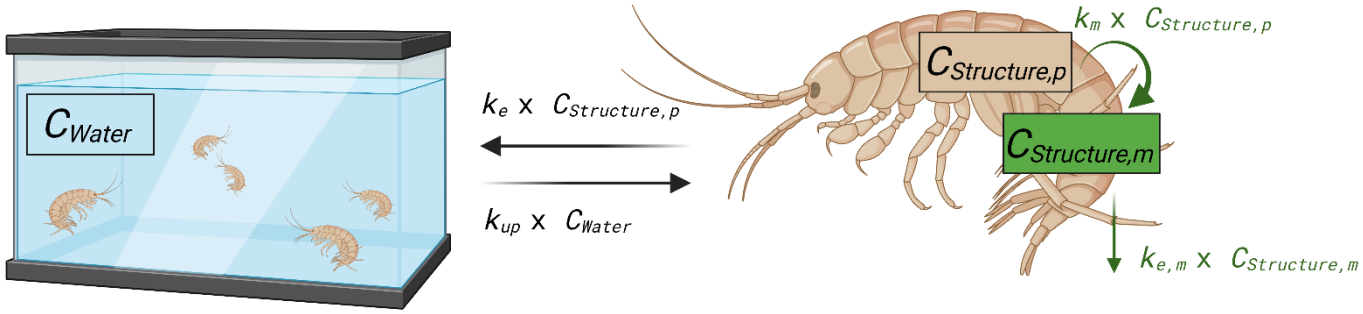
From laboratory to mesocosm: Temperature and species dependent toxicokinetics of the fluorinated herbicide Diflufenican (Submbitted, ES&T Toxicology)

Johannes Raths[1,2]*, Clemens Stäuble[1,2], Markus Hermann[1], Juliane Hollender[1,2]

Biotransformation
One-compartment model

$$BCF = \frac{C_{tissue,ss}}{C_{water,ss}} = \frac{k_u}{k_e + k_m}$$





$$\frac{dC_{structure,p}(t)}{dt} = k_u \cdot C_{water}(t) - k_e \cdot C_{structure,p}(t) - k_m \cdot C_{structure,p}(t)$$

$$\frac{dC_{structure,m}(t)}{dt} = k_m \cdot C_{structure,p}(t) - k_{e,m} \cdot C_{structure,m}(t)$$

Biotransformation
Mass Balance

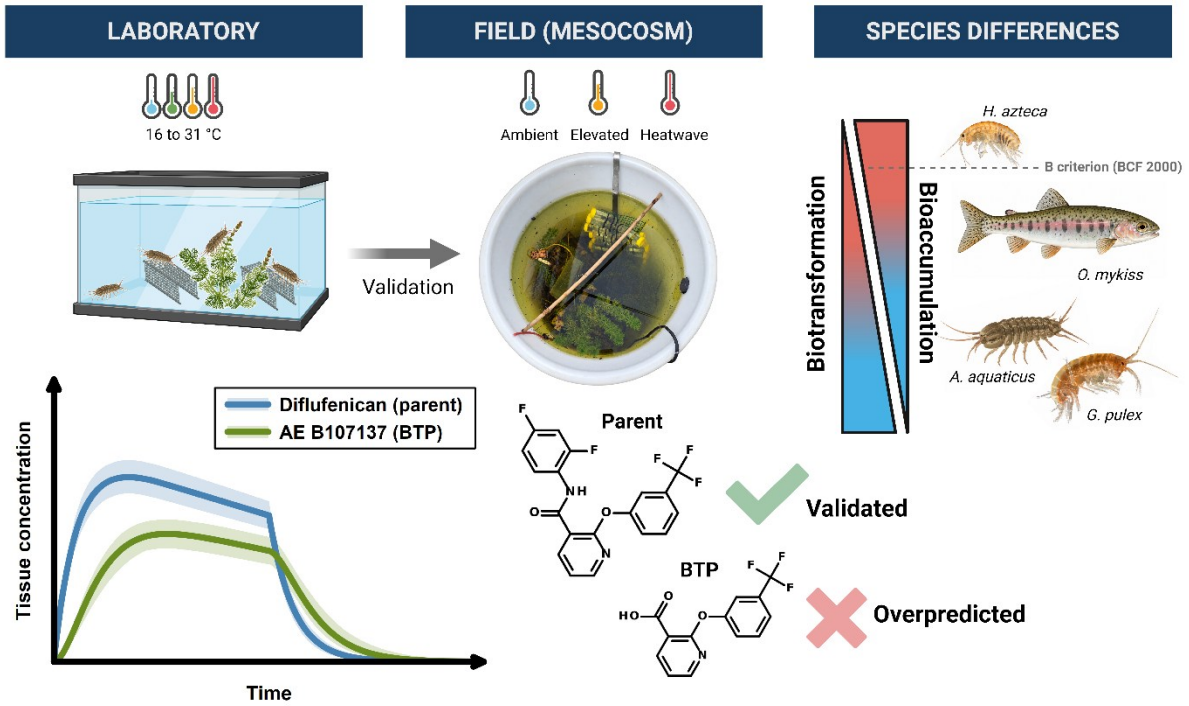
- The unit for biochemical processes is mol!
  - Biotransformation
  - Receptor interactions
  - …

1 g Diflufenican + 0.05 g H_2O → 0.72 g AE B107137 + 0.33 g 2,4-Difluoroaniline

Easier modeling using mol:
1 mol Diflufenican + 1 mol H_2O → 1 mol AE B107137 + 1 mol 2,4-Difluoroaniline

Dataset

1 Species, 1 compound



From laboratory to mesocosm: Temperature and species dependent toxicokinetics of the fluorinated herbicide Diflufenican

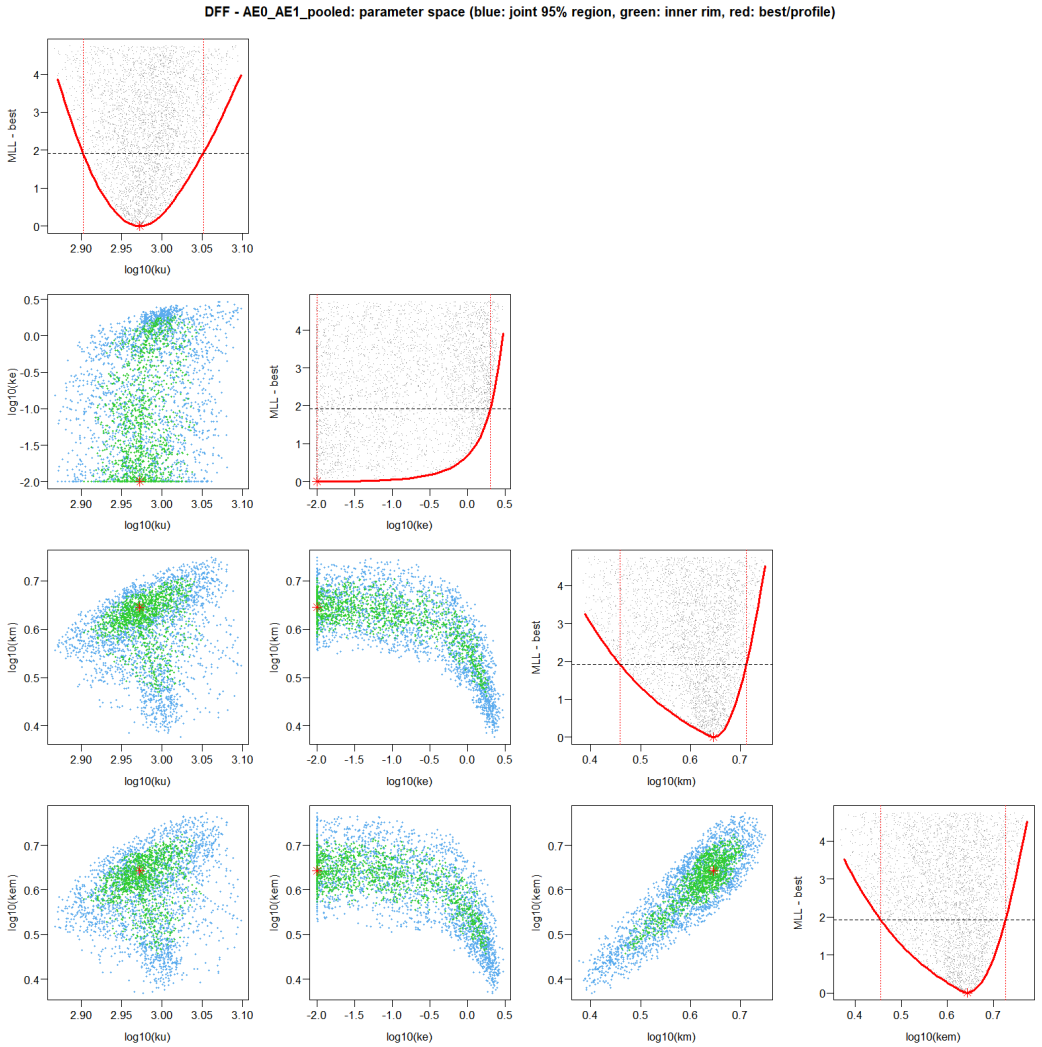
Johannes Raths[1,2]*, Clemens Stäuble[1,2], Markus Hermann[1], Juliane Hollender[1,2]

Task:
Run the model for different Experiments

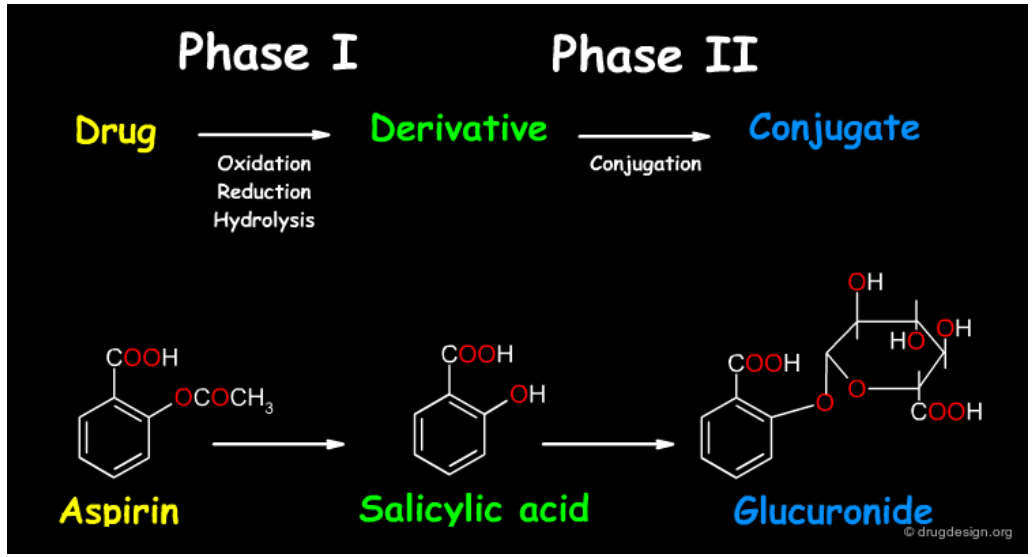
(i.e. AE0, AE1, AE0 + AE1)

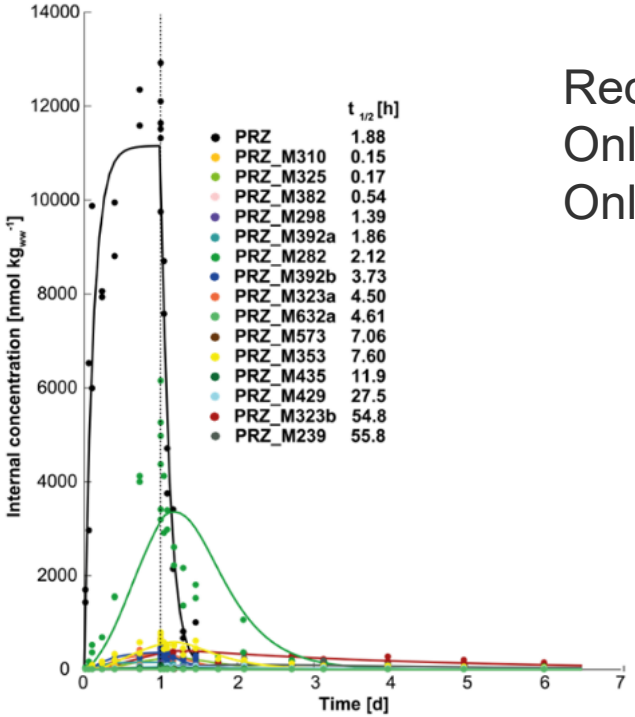
14:45-15:00 | Plenary discussion of Exercise 2

Uncertanties:
Overparametrization



More complex pathways
Phase I and Phase II





Reduction:
Only use BTPs > 5%
Only use Total BTPs

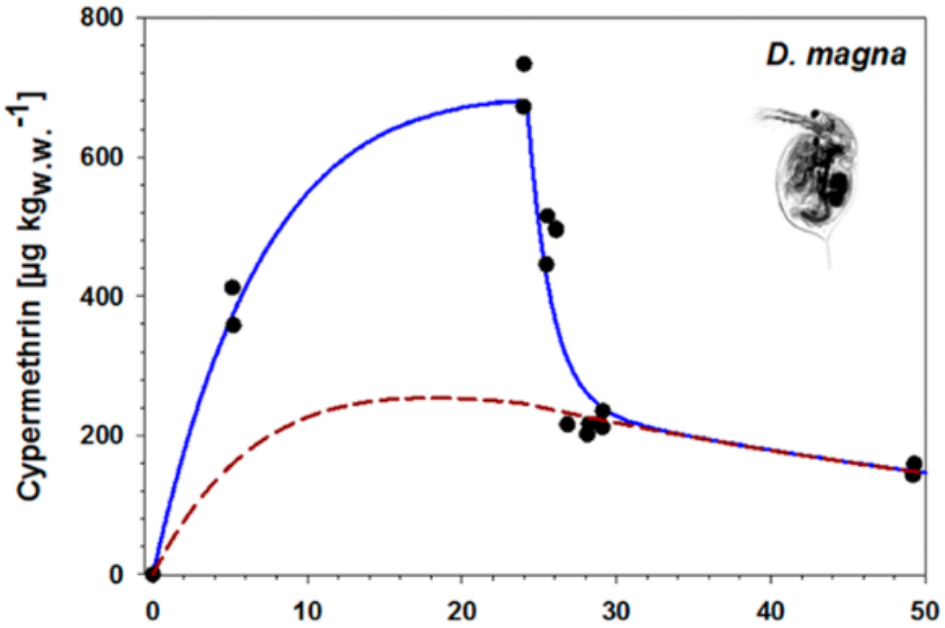
Rösch, A.; Anliker, S.; Hollender, J.
How Biotransformation Influences Toxicokinetics of Azole Fungicides in the Aquatic Invertebrate Gammarus Pulex.
Environ. Sci. Technol. **2016**, *50* (13), 7175–7188.
https://doi.org/10.1021/acs.est.6b01301

15:00–15:15 | Coffee break

15:15–15:45 | Exercise 3: Multiple compartment kinetics

Multiple compartment kinetics

Indications for multiple compartments (1)

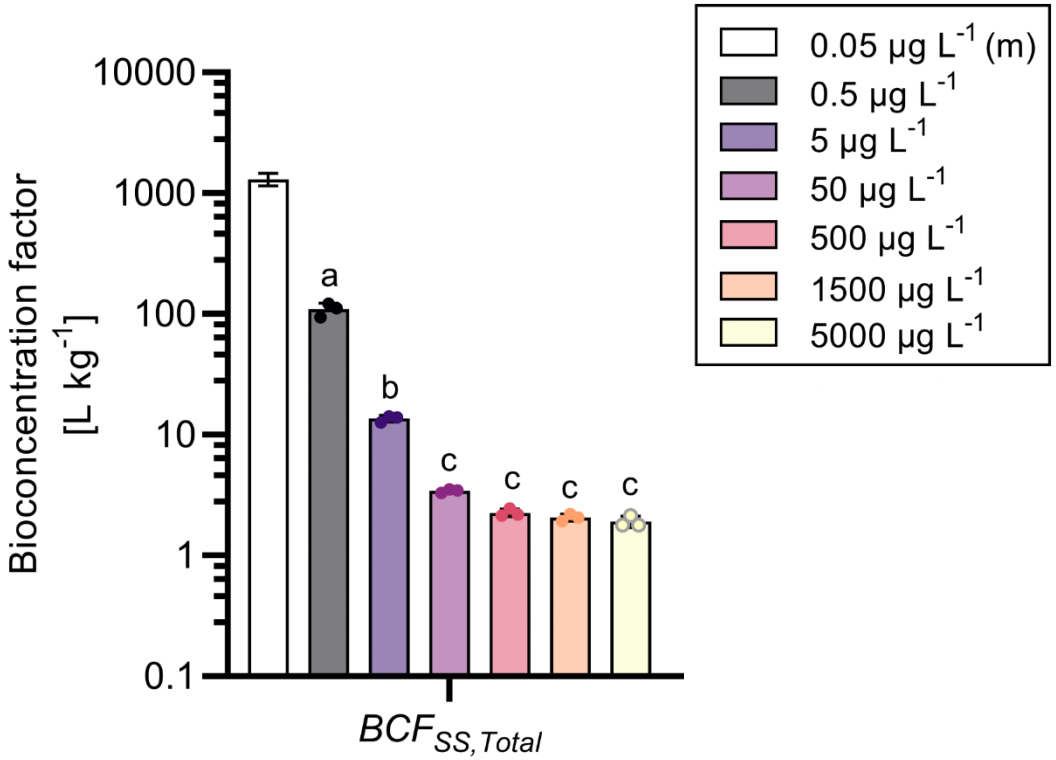


! Biphasic elimination

Dalhoff, K.; Hansen, A. M. B.; Rasmussen, J. J.; Focks, A.; Strobel, B. W.; Cedergreen, N. Linking Morphology, Toxicokinetic, and Toxicodynamic Traits of Aquatic Invertebrates to Pyrethroid Sensitivity. *Environ. Sci. Technol.* **2020**, *54* (9), 5687–5699. https://doi.org/10.1021/acs.est.0c00189

Multiple compartment kinetics

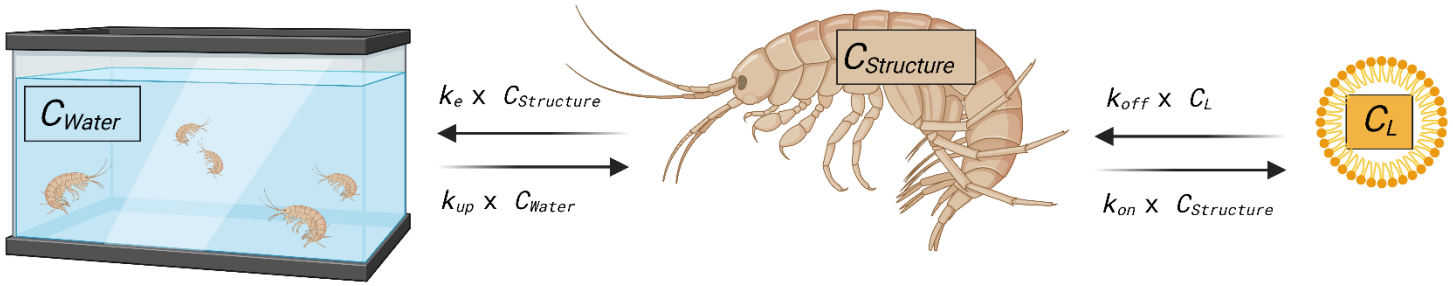
Indications for multiple compartments (2)



! Concentration dependent BCF (saturation)

Raths, J.; Schinz, L.; Mangold-Döring, A.; Hollender, J. Elimination Resistance: Characterizing Multi-Compartment Toxicokinetics of the Neonicotinoid Thiacloprid in the Amphipod Gammarus Pulex Using Bioconcentration and Receptor-Binding Assays. *Environ. Sci. Technol.* **2023**. https://doi.org/10.1021/acs.est.3c01891

Multiple compartment kinetics
Lipid partitioning



FMS
= Lipid content [%]

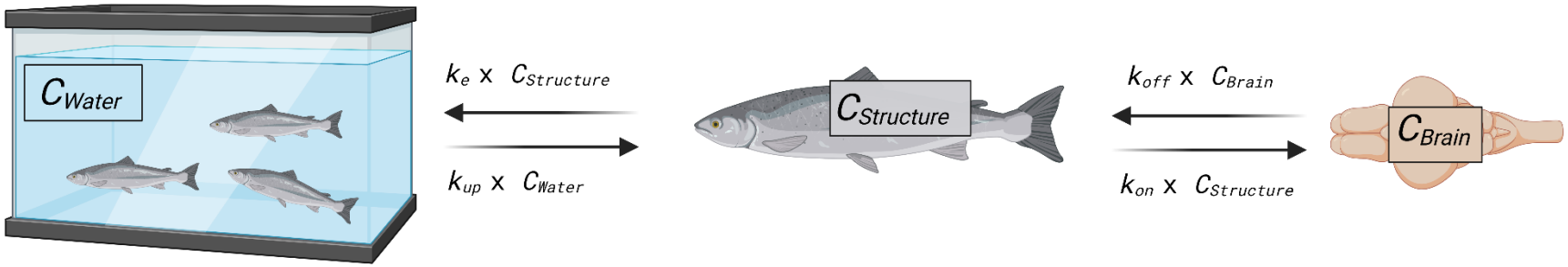
$$\frac{dC_{Structure}(t)}{dt} = k_u \cdot C_{water}(t) - k_e \cdot C_{Structure}(t) - k_{on} \cdot C_{Structure}(t) \cdot FMS + k_{off} \cdot C_{Lipid}(t) \cdot FMS$$

$$\frac{dC_{Lipid}(t)}{dt} = k_{on} \cdot C_{Structure}(t) - k_{off} \cdot C_{Lipid}(t)$$

Jager, T.; Øverjordet, I. B.; Nepstad, R.; Hansen, B. H.
Dynamic Links between Lipid Storage, Toxicokinetics and Mortality in a Marine Copepod Exposed to Dimethylnaphthalene.
Environ. Sci. Technol. **2017**, *51* (13), 7707–7713.
https://doi.org/10.1021/acs.est.7b02212

Multiple compartment kinetics

Brain partitioning



FMS
= Brain volume [%]

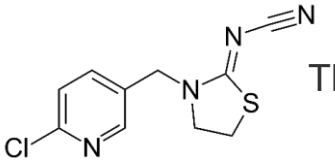
$$\frac{dC_{Structure}(t)}{dt} = k_u \cdot C_{water}(t) - k_e \cdot C_{Structure}(t) - k_{on} \cdot C_{Brain}(t) \cdot FMS$$

$$\frac{dC_{Lipid}(t)}{dt} = k_{on} \cdot C_{Structure}(t) - k_{off} \cdot C_{Brain}(t)$$

David, A.; Lange, A.; Tyler, C. R.; Hill, E. M. Concentrating Mixtures of Neuroactive Pharmaceuticals and Altered Neurotransmitter Levels in the Brain of Fish Exposed to a Wastewater Effluent. *Science of The Total Environment* **2018**, *621*, 782–790. https://doi.org/10.1016/j.scitotenv.2017.11.265

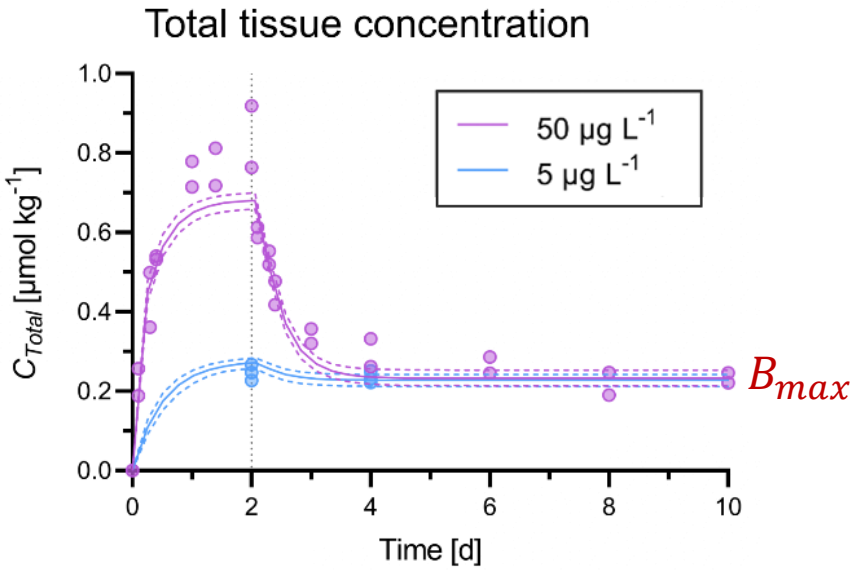
Multiple compartment kinetics

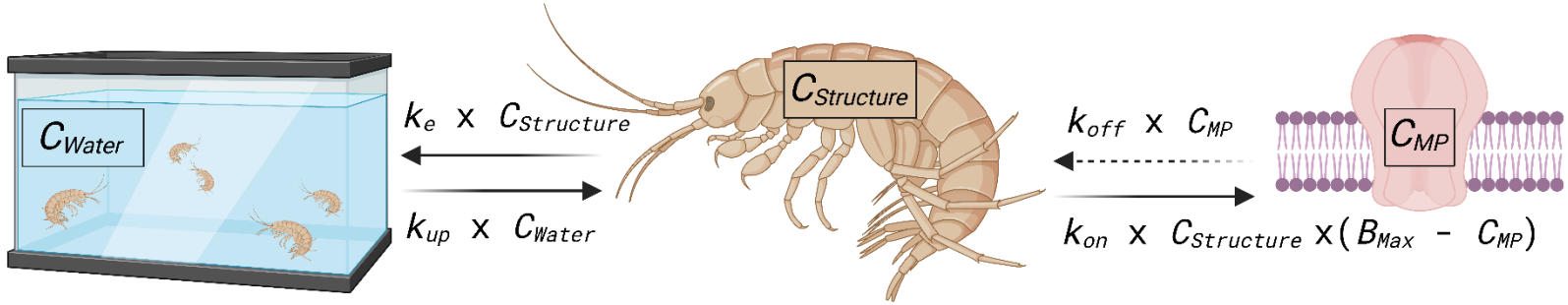
Receptor binding



Thiacloprid

FMS
= Membrane protein [%]





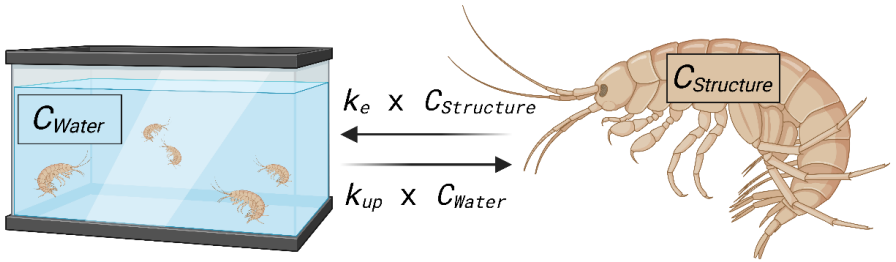
$$\frac{dC_{Structure}(t)}{dt} = k_u \cdot C_{water}(t) - k_e \cdot C_{Structure}(t) - \frac{dC_{MP}(t)}{dt} \cdot FMS$$

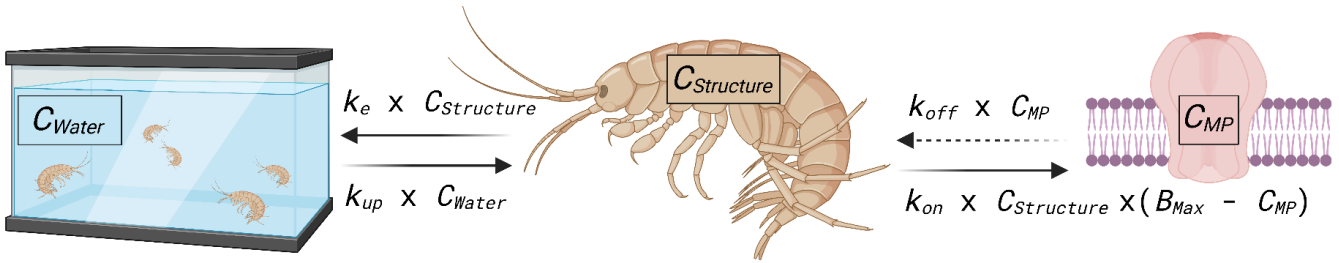
$$\frac{dC_{MP}(t)}{dt} = k_{on} \cdot C_{Structure}(t) \cdot \left(B_{max} - C_{MP}(t)\right)$$

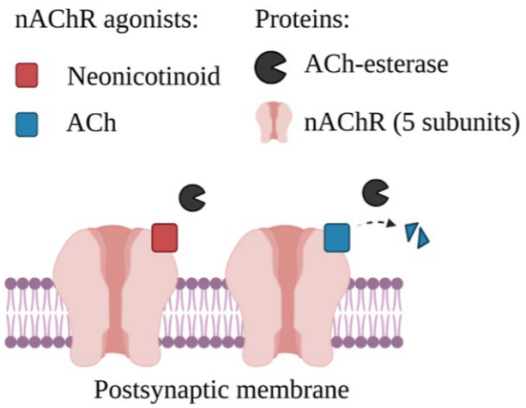
Raths, J.; Schinz, L.; Mangold-Döring, A.; Hollender, J. Elimination Resistance: Characterizing Multi-Compartment Toxicokinetics of the Neonicotinoid Thiacloprid in the Amphipod Gammarus Pulex Using Bioconcentration and Receptor-Binding Assays. *Environ. Sci. Technol.* **2023**. https://doi.org/10.1021/acs.est.3c01891

Dataset

Thiacloprid in *G. pulex*







Environmental Science & Technology
pubs.acs.org/est
Article

Elimination Resistance: Characterizing Multi-compartment Toxicokinetics of the Neonicotinoid Thiacloprid in the Amphipod *Gammarus pulex* Using Bioconcentration and Receptor-Binding Assays

Johannes Raths, Linda Schinz, Annika Mangold-Döring, and Juliane Hollender*

Cite This: *Environ. Sci. Technol.* 2023, 57, 8890–8901
Read Online

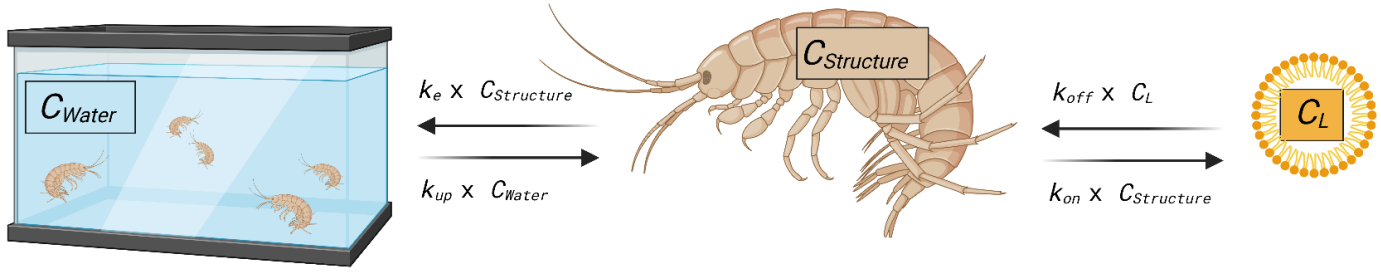
Task:
Compare 1- and 2-compartment model

Which data do you need?
Which model is better?

Run simplex first! – full model takes a while

Dataset:
Dimethylnaphthalene in Copepods

20 % lipid content 😋



Environmental Science & Technology
Article
pubs.acs.org/est

Dynamic Links between Lipid Storage, Toxicokinetics and Mortality in a Marine Copepod Exposed to Dimethylnaphthalene

Tjalling Jager,*,† Ida Beathe Øverjordet,‡ Raymond Nepstad,‡ and Bjørn Henrik Hansen‡

†DEBtox Research, NL-3731 DN, De Bilt, The Netherlands
‡SINTEF Ocean AS, Environmental Technology, N-7465 Trondheim, Norway

Task:
Fit the lipid model

Which data do you need?
What are limitations?

15:45-16:00 | Plenary discussion of Exercise 3

Model comparison

Akaike Information Criterion

$$\mathrm{AIC} = 2k - 2\ln(\hat{L})$$

k = number of estimated parameters (*including the residual variance σ²*)
$\hat{L}$ = maximized likelihood

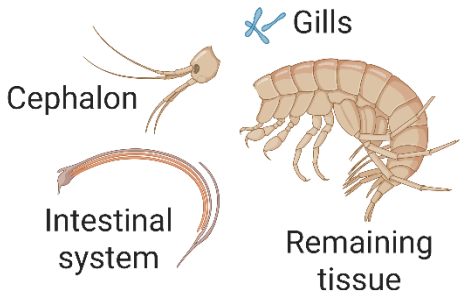
ΔAIC	Support for the model with the lower AIC
0–2	The two models are about equal → Keep simpler model
2–4	Grey zone
4–6	Considerably better
> 6	Substantialy better

Validation!

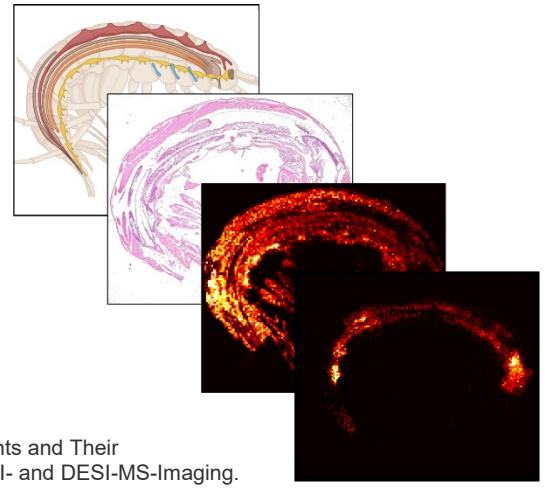
Measure concentrations in relevant compartments

- Brain
- Membrane protein
- Lipid partitioning
- Sorption to exoskeleton / skin

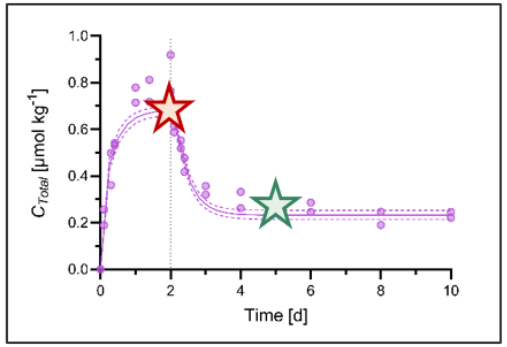
A Dissection

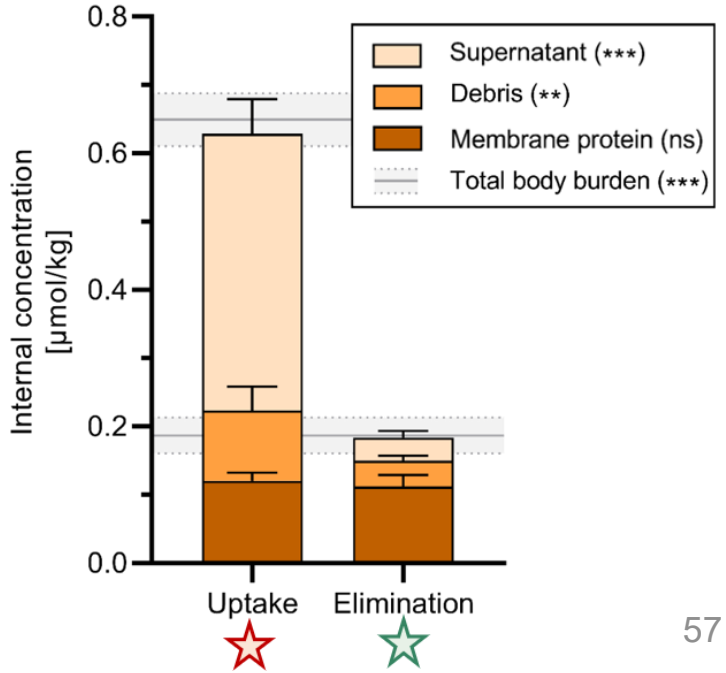


B Imaging



C Protein fractionating

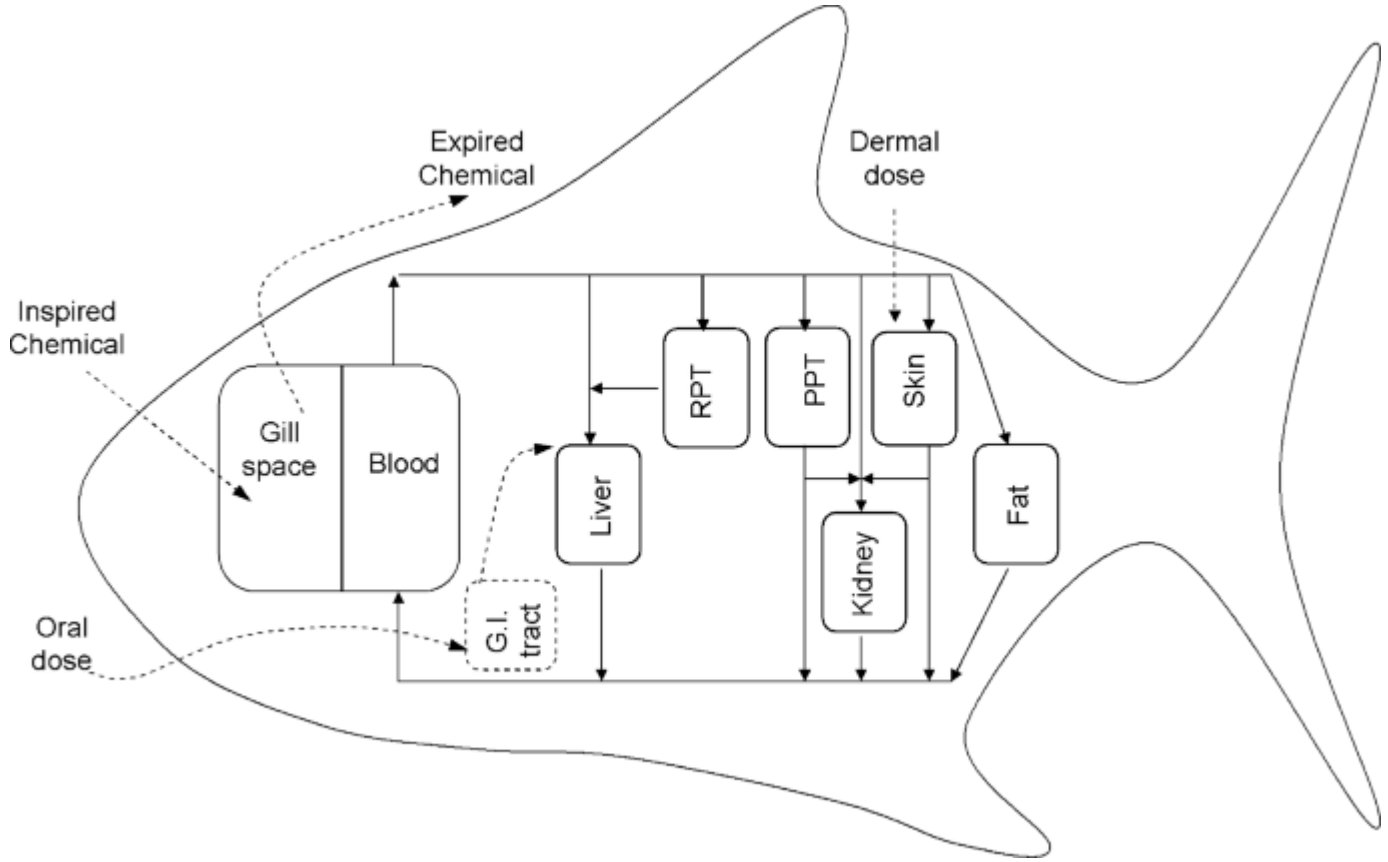




Raths, J.; Pinto, F. E.; Janfelt, C.; Hollender, J.
Elucidating the Spatial Distribution of Organic Contaminants and Their Biotransformation Products in Amphipod Tissue by MALDI- and DESI-MS-Imaging.
Ecotoxicology and Environmental Safety **2023**, *264*, 115468.
https://doi.org/10.1016/j.ecoenv.2023.115468

PBTK Models

But how many compartments are helpful?



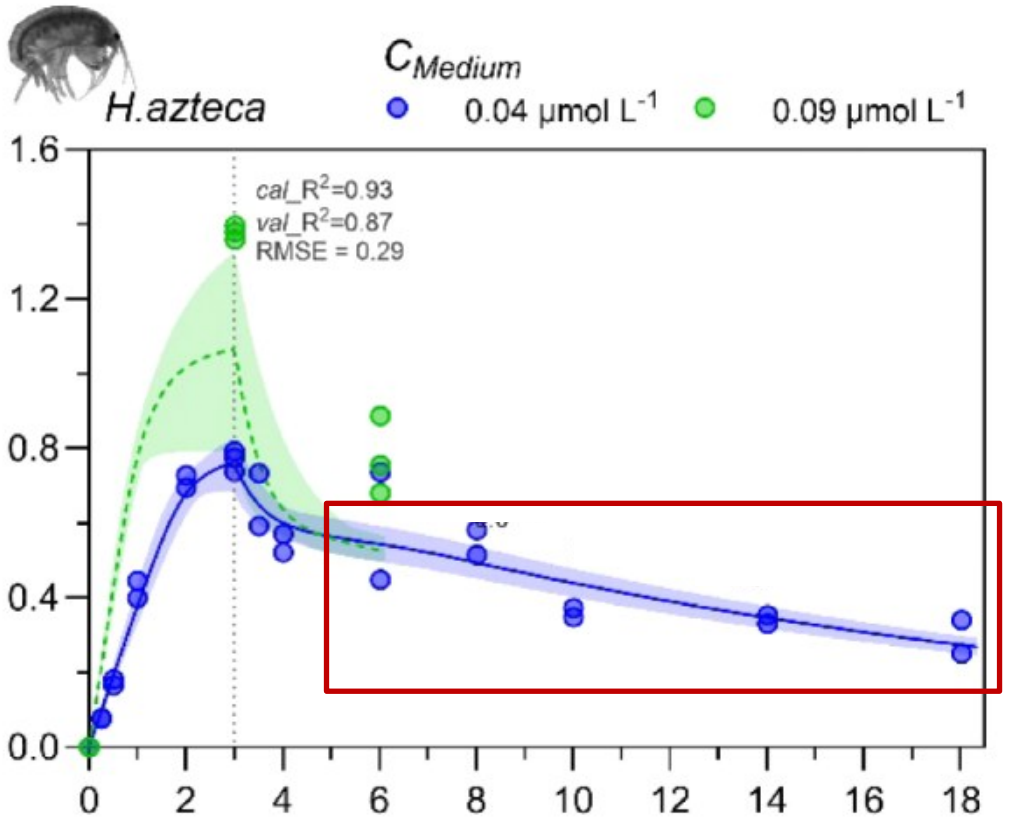
Krishnan, K., & Peyret, T. (2009). Physiologically based toxicokinetic (PBTK) modeling in ecotoxicology. In *Ecotoxicology modeling* (pp. 145-175). Boston, MA: Springer US.

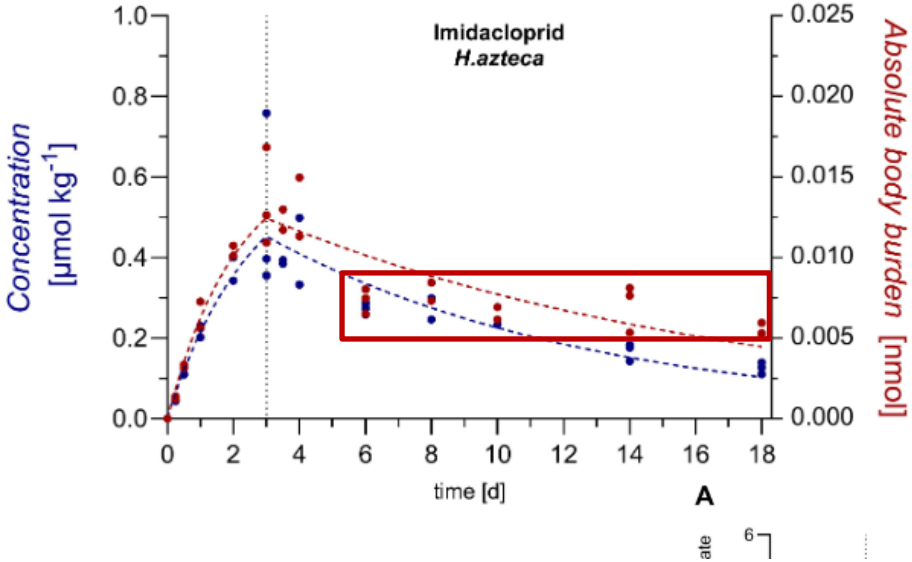
16:00–16:30 | Outlook: Dietary uptake, temperature effects and others

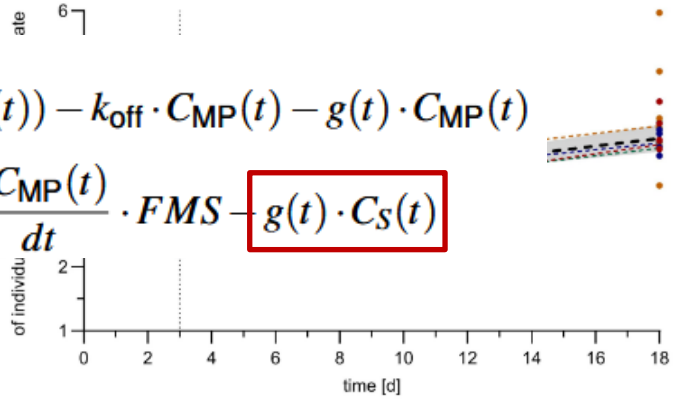
Growth Dilution

Real elimination or simply increase of biomass?

Von Au, C.; Maeda, N.; Raths, J.; Hollender, J. Receptor BindingDrives Toxicokinetic Processes: Insightsfrom Two Amphipod Species Exposed to Insecticides Targeting the NicotinicAcetylcholine Receptor. *Environmental Science & Technology* **2026** https://doi.org/10.1021/acs.est.6c08872





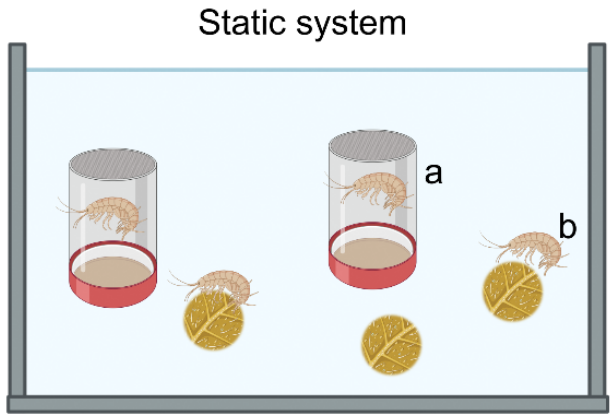


$$\frac{dC_{MP}(t)}{dt} = k_{on} \cdot C_S(t) \cdot (B_{max} - C_{MP}(t)) - k_{off} \cdot C_{MP}(t) - g(t) \cdot C_{MP}(t)$$

$$\frac{dC_S(t)}{dt} = k_u \cdot C_w(t) - k_e \cdot C_S(t) - \frac{dC_{MP}(t)}{dt} \cdot FMS - g(t) \cdot C_S(t)$$

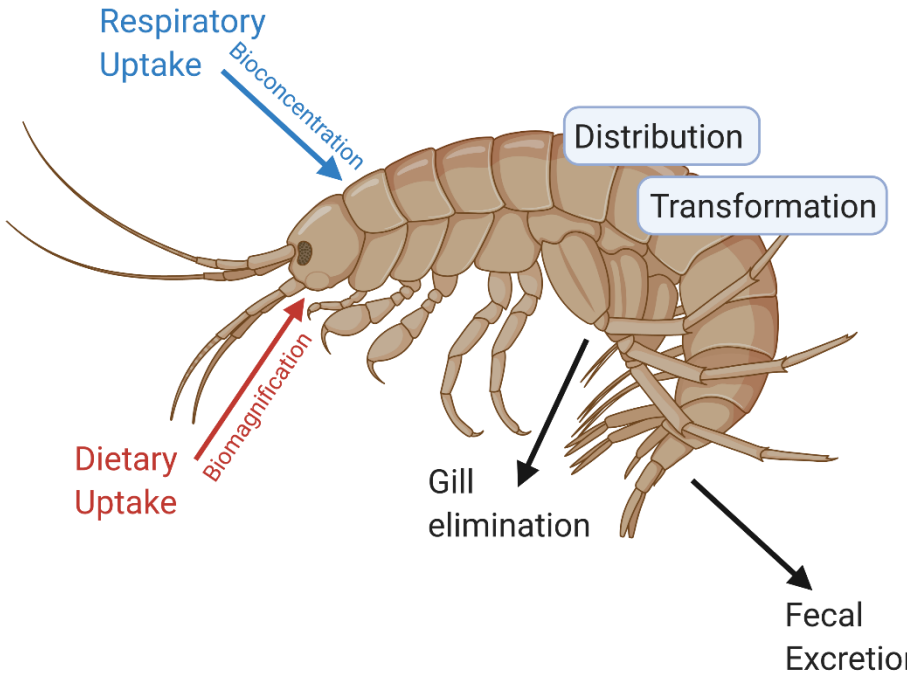
Dietary uptake

Adding feeding rate, assimilation efficiency and food concentration



Caged (a):
- No access to leaves
- Bioconcentration

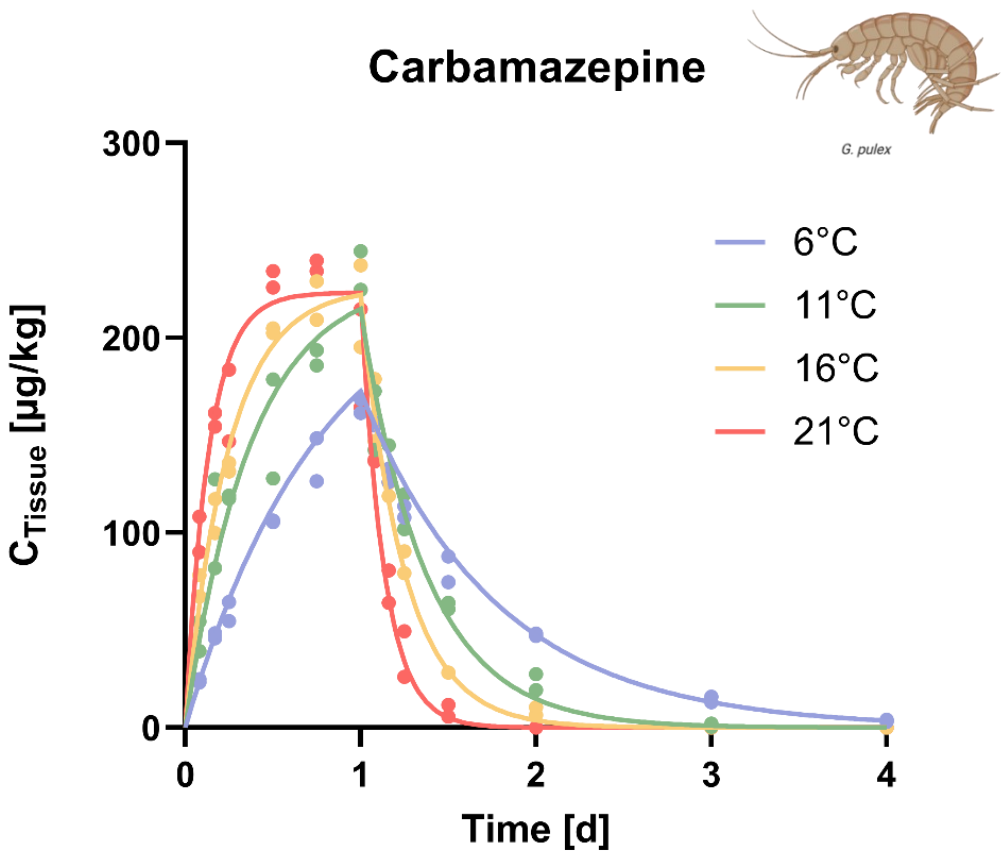
Uncaged (b):
- Access to leaves
- Bioconcentration and Biomagnification

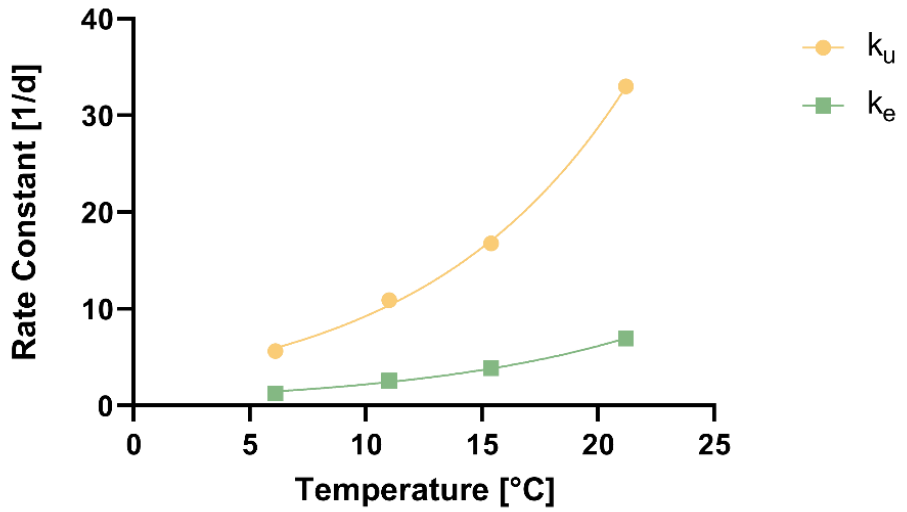


$$\frac{dC_{tissue}}{dt} = C_{water}\ k_u + C_{leaf}\ k_{feed}\ \alpha - C_{tissue}\ k_e$$

Temperature dependency

Using the Arrhenius Equation



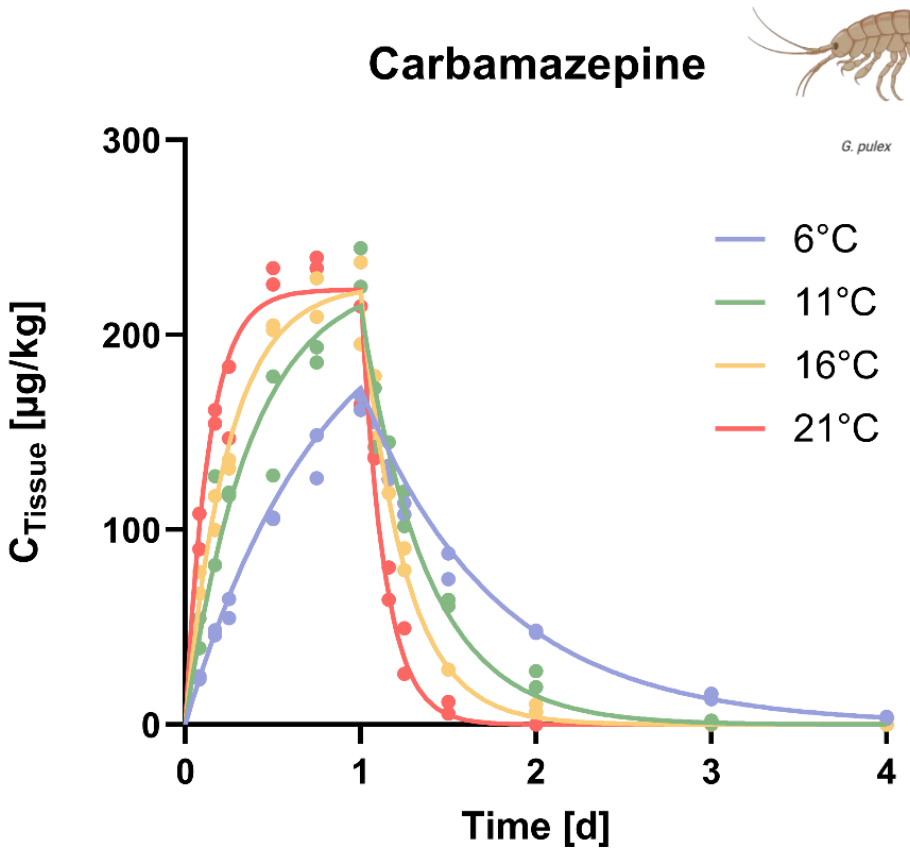


Temperature **exponentially affects** uptake and elimination kinetics

Do the rate constants increase proportionally?

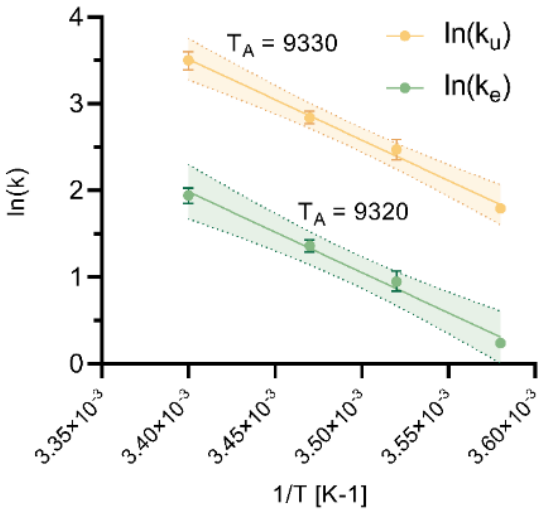
Temperature dependency

Using the Arrhenius Equation



Arrhenius Equation

(temperature dependent reaction rate)



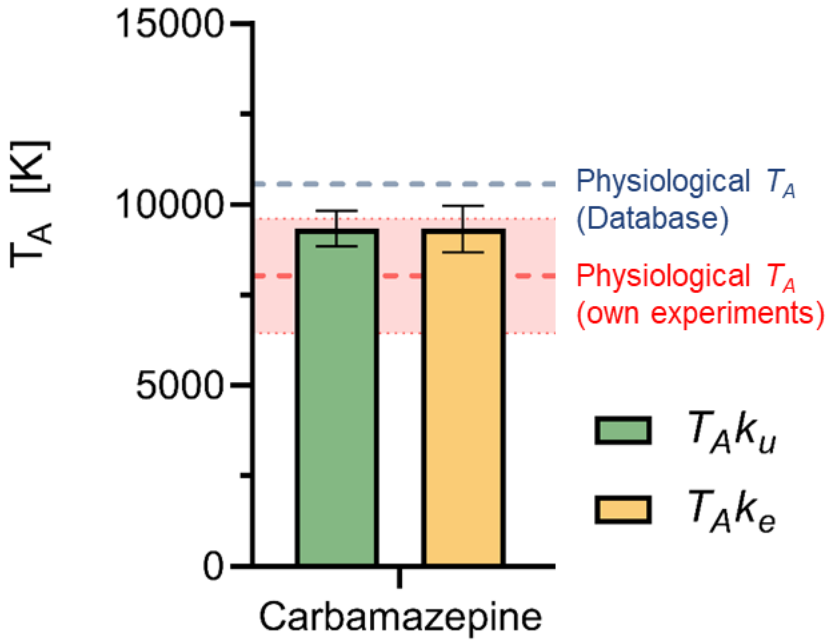
$$\ln k = -T_A \frac{1}{T} + \ln A$$

$T_A = Arrhenius\ Temperature$ $(activation\ energy)$

$A = frequency\ factor$ $(activation\ entropy)$

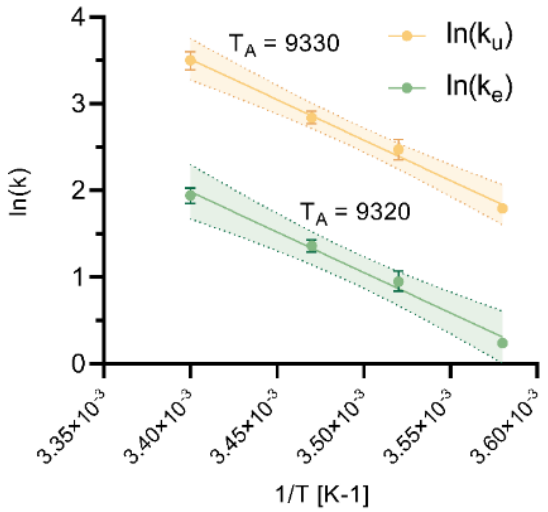
Temperature dependency

Using the Arrhenius Equation



Arrhenius Equation

(temperature dependent reaction rate)



$$\ln k = -T_A \frac{1}{T} + \ln A$$

$T_A = Arrhenius\ Temperature$ $(activation\ energy)$

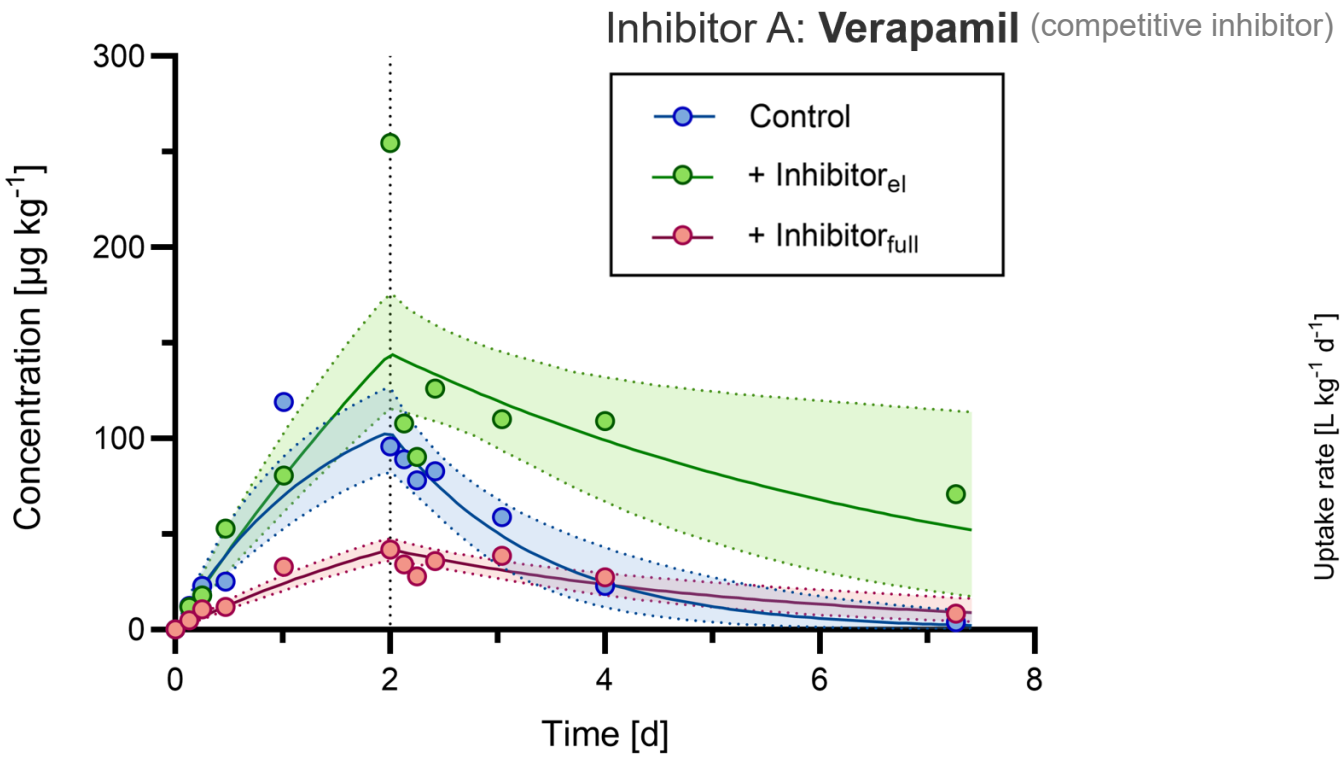
$A = frequency\ factor$ $(activation\ entropy)$

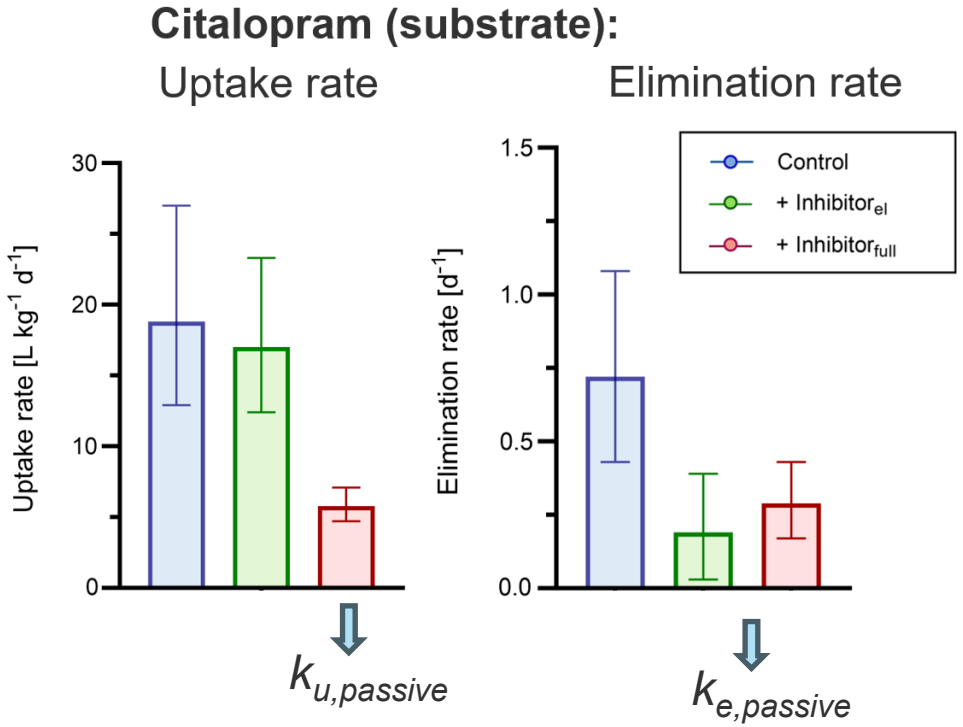
Add-my-Pet

https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/

Active Transport

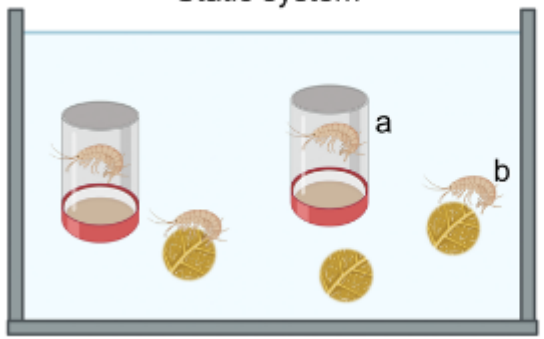
i.e. using two uptake rate constants (e.g. $k_{u,passive}$ and $k_{u,active}$)





Dataset
1 Species, 4 Pesticides

Static system



$$\frac{k_{feed}\ (kg_{dw}\ kg_{ww}^{-1}\ day^{-1})}{0.085}$$

Caged (a):
- No access to leaves
- Bioconcentration

Uncaged (b):
- Access to leaves
- Bioconcentration and Biomagnification

Task:
Fit the dietary uptake model

Which data do you need?
Which parameters are fixed/fitted?
How to compare BCF and BMF?

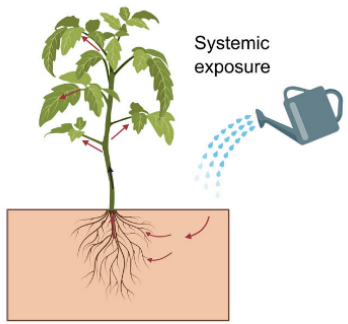
Environmental Toxicology and Chemistry—Volume 42, Number 9—pp. 1993–2006, 2023
Received: 12 January 2023 | Revised: 1 March 2023 | Accepted: 17 March 2023

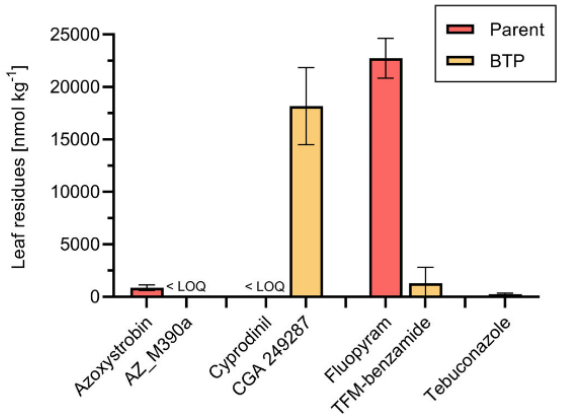
Environmental Toxicology

OPEN DATA OPEN MATERIALS

Importance of Dietary Uptake for in Situ Bioaccumulation of Systemic Fungicides Using *Gammarus pulex* as a Model Organism

Johannes Raths,[a,b] Jacob Schnurr,[c] Mirco Bundschuh,[c,d] Fernanda E. Pinto,[e] Christian Janfelt,[e] and Juliane Hollender[a,b,*]





16:30–17:00 | Final discussion and Wrap-up

TK Databases

https://mosaic.univ-lyon1.fr/tkdb/

Accumulation-depuration data collection in support of toxicokinetic modelling

Aude Ratier, Sandrine Charles & Miléna Kaag

2024-08-30

Introduction

This database contains a wide collection of raw data sets extracted from the scientific literature in support of TK modelling which can be performed with the MOSAIC web application (https://mosaic.univ-lyon1.fr/bioacc/).

More details on the database are given in the companion paper.

All data sets are summarized by:

- the name of the file of the data set (directly downloadable)	- the total number of data in the dataset (number of replicate(s))
- the organism tested (genus)	- the kinetic bioaccumulation factor median value with its 95% credible intervals
- the category of the organism (*e.g.* aquatic, terrestrial, etc.)	- the report in .pdf (directly downloadable)
- the substance tested	- the reference or source of the data set
- the time at the end of the accumulation phase (t_{acc})	- possible comments added (*e.g.* lipid fraction, growth, biotransformation)
- the exposure route tested and for which we have data (*e.g.* water, sediment, food, pore water)	

Your contribution

If you want to contribute to this database in sharing your raw TK data, please contact us at sandrine.charles@univ-lyon1.fr.

Download all data sets

You can access to all data here : https://zenodo.org/badge/latestdoi/359723940

Databases / Prediction Tools

https://www.epa.gov/comptox-tools/high-throughput-toxicokinetics-httk-r-package



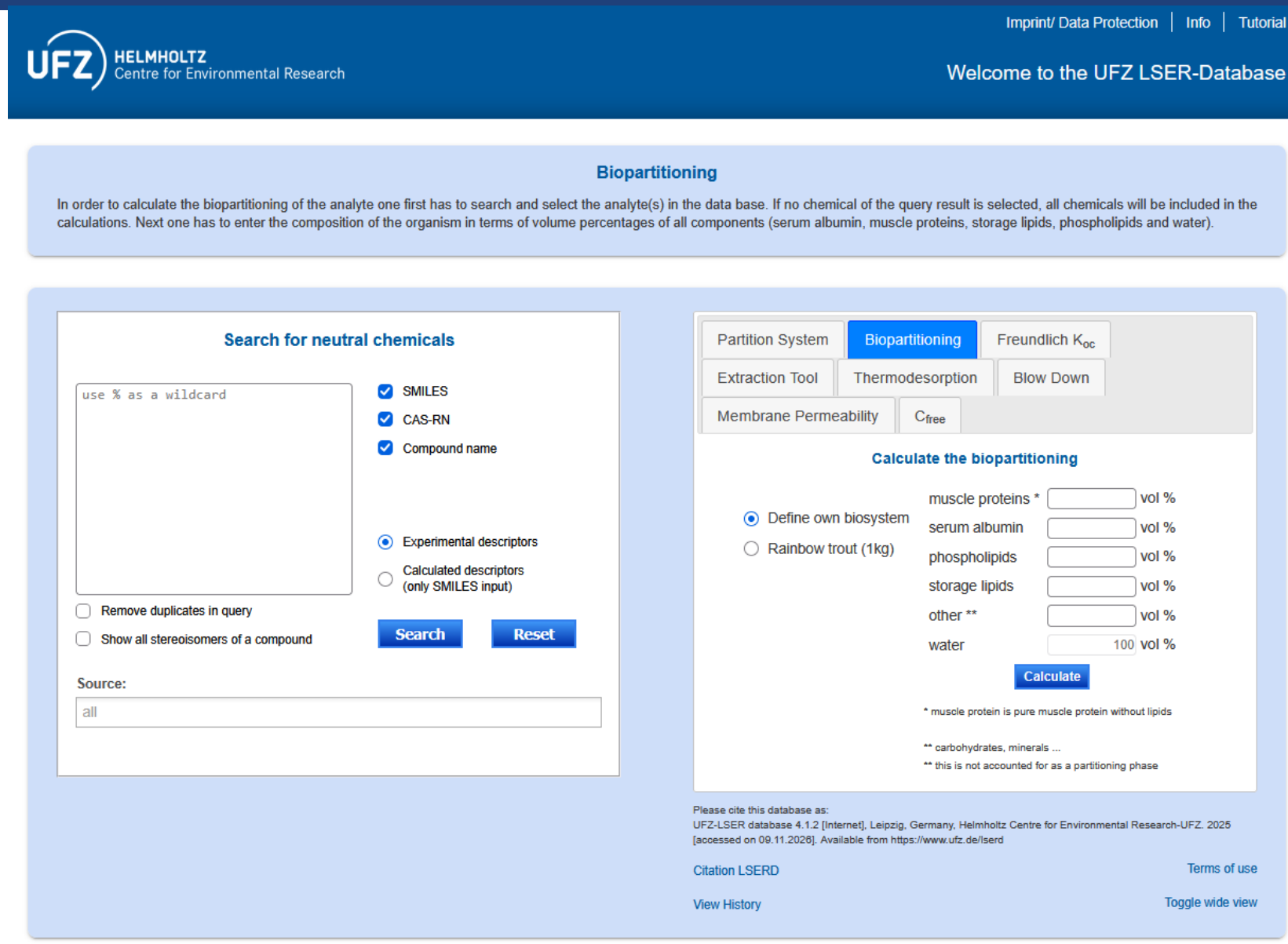
Prediction Tools

https://web.app.ufz.de/compbc/lserd/public/start/

Endo, S.; Escher, B. I.; Goss, K.-U. Capacities of Membrane Lipids to Accumulate Neutral Organic Chemicals. *Environ. Sci. Technol.* **2011**, *45* (14), 5912–5921. https://doi.org/10.1021/es200855w

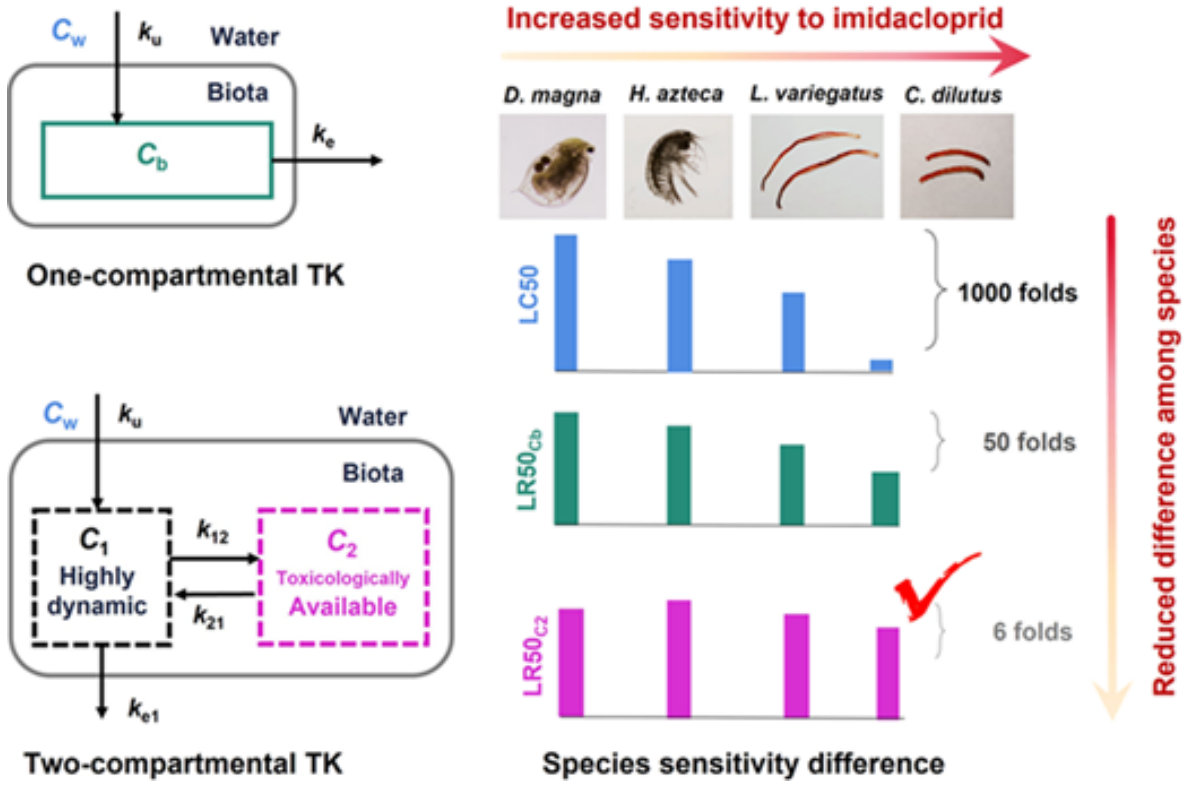
Endo, S.; Bauerfeind, J.; Goss, K.-U. Partitioning of Neutral Organic Compounds to Structural Proteins. *Environ. Sci. Technol.* **2012**, *46* (22), 12697–12703. https://doi.org/10.1021/es303379y

Endo, S.; Brown, T. N.; Goss, K.-U. General Model for Estimating Partition Coefficients to Organisms and Their Tissues Using the Biological Compositions and Polyparameter Linear Free Energy Relationships. *Environ. Sci. Technol.* **2013**, *47* (12), 6630–6639. https://doi.org/10.1021/es401772m



Linking TK to TD

Internal concentration leads to effect

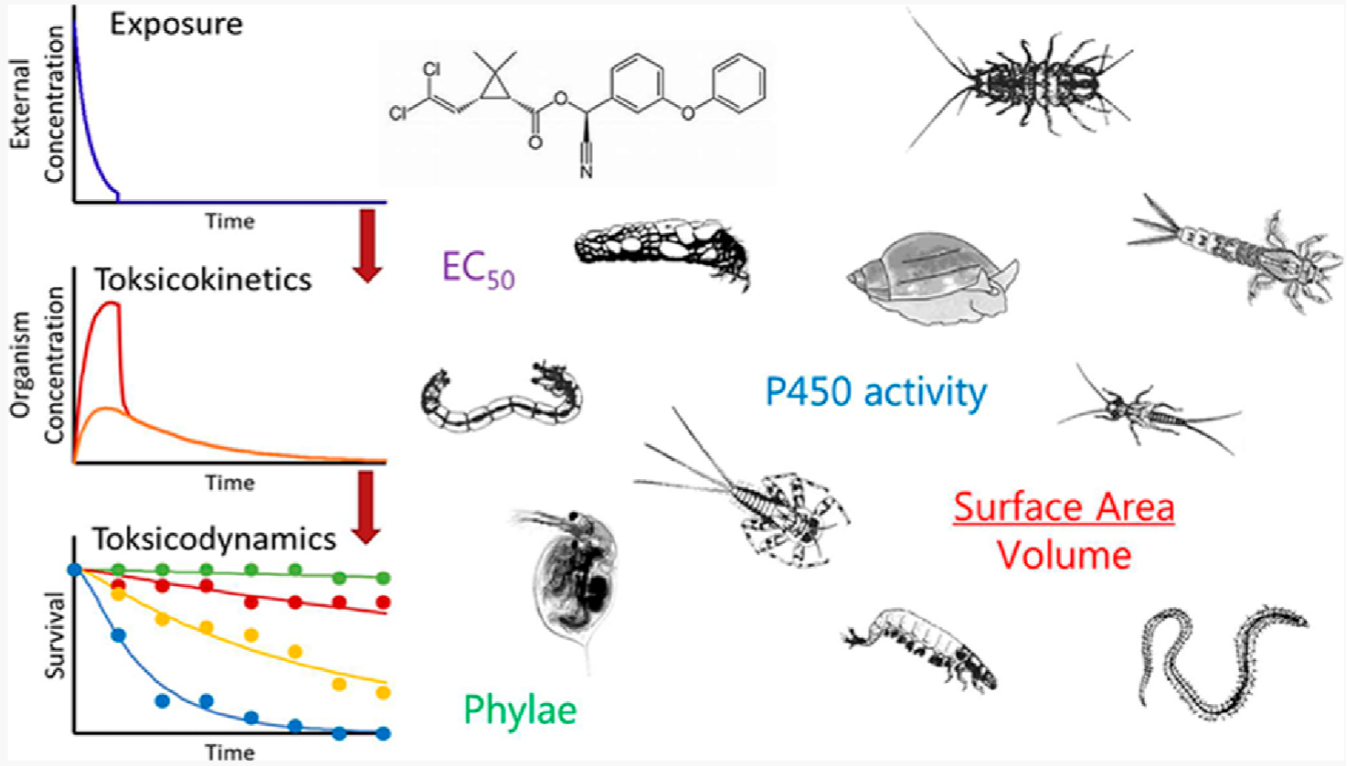


→ Assessing species differences
→ Explaining time dependent toxicity

Su, H.; Zhang, Q.; Huang, K.; Wang, W.-X.; Li, H.; Huang, Z.; Cheng, F.; You, J. Two-Compartmental Toxicokinetic Model Predicts Interspecies Sensitivity Variation of Imidacloprid to Aquatic Invertebrates. *Environ. Sci. Technol.* **2023**, *57* (29), 10532–10541. https://doi.org/10.1021/acs.est.3c01646

Linking TK to TD

Internal concentration leads to effect



→ Assessing species differences
→ Explaining time dependent toxicity

Dalhoff, K.; Hansen, A. M. B.; Rasmussen, J. J.; Focks, A.; Strobel, B. W.; Cedergreen, N.
Linking Morphology, Toxicokinetic, and Toxicodynamic Traits of Aquatic Invertebrates to Pyrethroid Sensitivity.
Environ. Sci. Technol. **2020**, *54* (9), 5687–5699.
https://doi.org/10.1021/acs.est.0c00189

Feedback

Questions for the plenum:

- How to improve the model / output
- → Accessibility, high throughput,

https://docs.google.com/forms/d/e/1FAIpQLScKwdTY8gJgJVbarcyWU4sAdq5NGjjyMyFI6YPSX4f9JtpBPw/viewform?usp=dialog